Joint Beamformer Design and Power Allocation Method for Hybrid RF-VLCP System

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***Abstract*—In this article, a hybrid radio frequency-visible light communication and positioning (RF-VLCP) system is designed, which can support high-data rate communication and high accuracy positioning with good energy efficiency (EE) performance. The hybrid system uses two links for downlink communication, namely, radio frequency (RF) and visible light communication (VLC) links, and employs visible light positioning (VLP) technology for positioning. Furthermore, an optimization problem is developed to allocate power for VLC and VLP links and to design beamformer for the RF transmitter. By doing so, the EE of the hybrid system is maximized while the Cramer–Rao Lower bound (CRLB) of the positioning error and the minimum data rate of the communication are guaranteed. A two-step algorithm is proposed to tackle the formulated optimization problem, which first determines the power allocation of the VLP signal and then obtains the power allocation of the VLC signal and the beamformer of the RF transmitter. Numerical results demonstrate the advantages of the proposed two-step algorithm over the existing algorithm in terms of computation speed. In addition, the EE performance of the hybrid system is evaluated under different data rate and positioning accuracy requirements. Besides, we also show that the hybrid RF-VLCP system is more energy efficient compared to standalone RF and VLP technologies.**

***Index Terms*—Beamforming, energy efficiency (EE), positioning accuracy, power allocation, visible light communication and positioning (VLCP).**

I. Introduction

NOWADAYS, the demand for high data rate communication is rapidly increasing due to the popularity of multimedia applications. Radio frequency (RF)-based communication is restricted by the limited spectrum resources and may not be able to meet the ever-increasing traffic demand. More recently, the visible light communication (VLC) is developed as an alternative technique. The VLC system is operated on a wide unlicensed spectrum, which can be reused in different rooms since visible light cannot pass through the walls. In addition, it is more secure and energy efficient than the RF system. However, despite these advantages, VLC also exhibits some limitations, such as small coverage area, low mobility, and limited transmitted power [1]. Since the RF and VLC systems are somewhat complementary to each other, the concept of a hybrid RF-VLC system is developed to support communication service by both RF and VLC links, which combines the positive features of the two technologies and overcomes the limitations of the individual links. So far, a large amount of literature has discussed various issues of hybrid RF-VLC systems, including but not limited to load balance [2], [3], [4], coverage and rate analysis [5], [6], [7], and energy efficiency (EE) performance [8], [9], [10], [11].

On the other hand, indoor positioning is also attractive and urgent in many scenarios. The global positioning system (GPS), which has been widely used for outdoor positioning, may not be sufficiently accurate when used indoors, due to the attenuation of GPS signals caused by the interruptions of ceilings, walls, and other obstacles [12]. For this reason, indoor RF signals, such as WiFi and Bluetooth, were first considered for indoor positioning [13]. However, the performance of RF-based positioning can be significantly affected by multipath interference which is particularly severe in indoor environments. To overcome this, researchers then began to investigate the visible light positioning (VLP) technology [14], [15], [16]. Compared to the RF signal, the optical signal is less affected by multipath interference, thus the VLP can be more accurate than RF-based positioning [17]. Up to now, the VLP technology has been well investigated based on various positioning measurements, including received signal strength (RSS) [18], [19], [20], time of arrival (TOA) [21], time difference of arrival (TDOA) [22], [23], [24], [25], and angle of arrival (AOA) [26]. Moreover, since simultaneous communication and positioning is desired in some indoor scenarios (hospital, supermarket, etc.), the integrated VLC and VLP (VLCP) system has also been attracting attention. For example, Lin et al. [27] proposed an orthogonal frequency division multiplexing (OFDM)-based VLCP system, in which the subcarriers having the maximum signal strength are selected for VLP and the remaining subcarriers are used

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for VLC. Also considering the OFDM-based VLCP system, Yang et al. [28] discussed a joint coordinated subcarrier and power allocation approach to optimize and guarantee the performance of communication and positioning.

Up to now, there has been a rich discussion on hybrid RF-VLC systems and VLCP systems. The related technologies have also been well promoted in practical applications. However, these two systems can be incompetent when high quality communication and positioning are both required, because the hybrid RF-VLC system does not support the positioning service and the achievable data rate of the VLCP system is limited with a single VLC link. To this end, we consider a hybrid radio frequency-visible light communication and positioning (RF-VLCP) system, which inherits the advanced characteristics of the hybrid RF-VLC system and the VLCP system. The main content and contribution are summarized as follows.

1) We design an indoor hybrid RF-VLCP system, where the RF and VLCP systems use OFDM and direct current (DC) biased optical OFDM (DCO-OFDM) for signal modulation, respectively. The hybrid RF-VLCP system fills the deficiency of the existing VLCP system and RF-VLC system, as it can support high data rate communication with two links (i.e., RF and VLC), while achieving high accuracy positioning with the VLP technique.
2) We formulate an optimization problem to address the power allocation of the VLCP transmitter and the beamformer design of the RF transmitter for the hybrid RF-VLCP system. The formulated optimization problem aims to maximize the EE of the hybrid system while controlling the Cramer–Rao Lower bound (CRLB) of the positioning error and guaranteeing the minimum communication data rate. In particular, when accounting for the data rate of the VLC link, we consider the effect of clipping operation, which is necessary for DCO-OFDM modulated signal but is rarely considered in the related works regarding to VLCP or hybrid RF-VLC system.
3) We propose a two-step algorithm to solve the formulated optimization problem, which first determines the power allocation of the VLP signal, and then jointly optimizes the power allocation of the VLC signal and the RF signal beamformer. More specifically, the subproblem presented in the first step is solved within the framework of the alternating direction method of multipliers (ADMMs), and the subproblem formulated in the second step is solved by a Dinkelbach-based algorithm.
4) We show that the proposed two-step algorithm consumes much less computation time than the existing DSAC algorithm [29], while achieving almost the same optimization result. Besides, we evaluate the performance of the hybrid RF-VLCP system under different positioning accuracy requirements and data rate requirements. We also demonstrate that for simultaneous communication and positioning, the proposed hybrid system is more energy efficient compared to the standalone RF and VLP technologies.

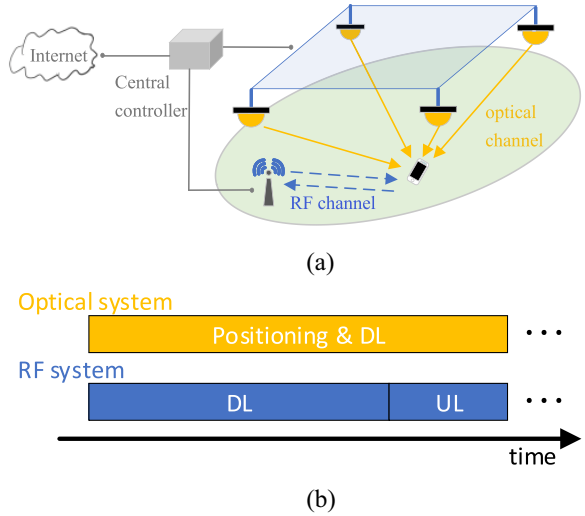

Fig. 1. Typical example of an indoor hybrid RF-VLCP system. (a) Graphical illustration: the hybrid RF-VLCP system employs a set of LEDs and an RF device to provide communication and positioning services. (b) Operating mechanism: the optical channel provides DL communication and positioning services continuously, while the RF channel provides DL and UL communication services alternately.

Notations: Standard lower case letters represent the constants or variables. Boldface letters in lower case and upper case denote the column vectors and matrices, respectively. $\mathbb{R}^N$ and $\mathbb{C}^N$ denote the N-dimensional real and complex vector spaces, respectively. $\otimes$ is the Kronecker product. $(\cdot)^{\mathrm{T}}$, $(\cdot)^*$ and $(\cdot)^H$, respectively, denote transpose operation, complex conjugate operation and complex conjugate transpose operation. For a variable x, $|x|$ and $\mathrm{Re}\{x\}$, respectively, denote the modulus and real part of it. For a vector $\mathbf{x}$, $\|\mathbf{x}\|$ denotes the Euclidean norm and $\mathbf{x}[m]$ denotes its mth element. For a matrix $\mathbf{X}$, $|\mathbf{X}|$ denotes the determinant of the matrix and $\mathbf{X}[m, n]$ represents the mth row and the nth column element. Besides, $\mathbf{I}$ stands for the identity matrix and $\mathbf{0}$ stands for a matrix whose elements are all zero.

II. System Structure

This article considers an indoor hybrid RF-VLCP system, as shown in Fig. 1. A group of M LEDs is mounted on the ceiling of the room, and an RF device equipped with N antennas is placed somewhere inside the room. The hybrid system is supposed to provide positioning and communication services for a user, which is equipped with a single photodetector (PD) and a single RF antenna. The operating mechanism of the hybrid system is explained by Fig. 1(b). That is, the positioning service is assigned to the LEDs, since VLP can be more accurate in an indoor environment compared to RF-based positioning [17]. The downlink (DL) communication is jointly supported by VLC and RF links, while the uplink (UL) communication can only be implemented through the RF link. As a result, the LEDs form a VLCP transmitter to continuously emit VLC and VLP signals, and the RF device transmits or receives signals for DL or UL communication in a time-division multiplexed manner.

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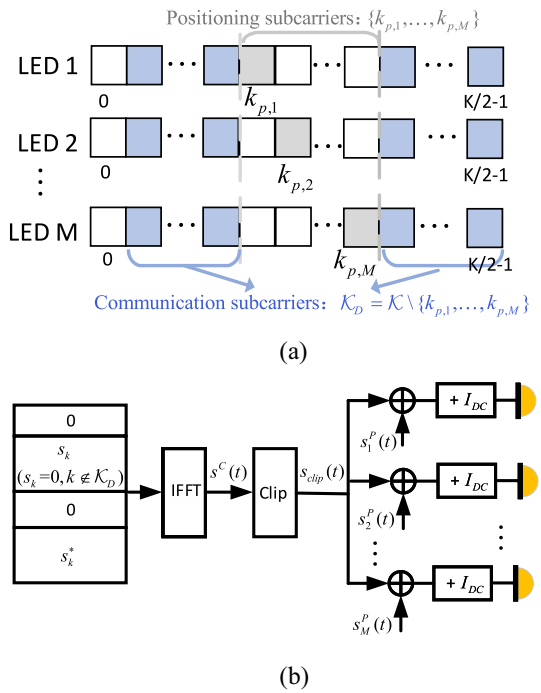

Fig. 2. Detailed arrangement of VLCP system. (a) Subcarrier allocation of the VLCP system. (b) Generation of the transmitted VLCP signal.

In this article, we focus on the joint DL and positioning operation in Fig. 1(b) and attempt to optimize the EE performance of the hybrid RF-VLCP system while guaranteeing the quality of DL communication and positioning. For this purpose, we formulate an optimization problem to jointly design the transmit beamformer of the RF signal and the power allocation of VLC and VLP signals. For practical implementation, the optimization issue can be processed by a central controller, which is typically used to relay data traffic from Internet to the VLCP transmitter and the RF transmitter. In this work, it also collects necessary information required by optimization issue, then tackles the joint optimization problem and sends the transmit scheme to two transmitters, respectively.

A. Signal Model of the VLCP System

The VLCP transmitter adopts the DCO-OFDM technique for signal modulation, which is a prevailing scheme in VLC/VLCP system due to its simple implementation and high spectral efficiency [30]. Then, for a VLCP system with K subcarriers, the symbols transmitted on the subcarriers should be Hermitian symmetric and the subcarriers at positions $\{0, K/2\}$ should be set to zero, so that the transmitted signal is real-valued. This means that only $K/2-1$ subcarriers (indexed by $\mathcal{K}=\{1,\ldots,K/2-1\}$ below) can be arranged for allocating VLC and VLP signals, and the detailed arrangement is illustrated in Fig. 2.

Specifically, each LED will designate a dedicate subcarrier to transmit the VLP signal. Suppose that the mth LED designates the $k_{p,m}$th subcarrier, the VLP signal it transmits is given as

$$\begin{aligned} s_m^P(t) &= \exp\big(j2\pi k_{p,m}\Delta f_v t\big) + \exp\big(-j2\pi k_{p,m}\Delta f_v t\big) \\ &= 2\cos\big(2\pi k_{p,m}\Delta f_v t\big) \end{aligned} \tag{1}$$

where Δf_v is the frequency spacing of adjacent subcarriers. Note that the $k_{p,m}$th subcarrier is not available for the other LEDs, since it is dedicated to the mth LED.

Except for M positioning subcarriers, the remaining subcarriers are used for data transmission. The index set of data subcarriers is denoted as $\mathcal{K}_D=\mathcal{K}\backslash\{k_{p,1},\ldots,k_{p,M}\}$. Different from positioning subcarriers, each data subcarrier is available to all the LEDs, and the LEDs will transmit the same symbol on each data subcarrier. Therefore, denoting the symbol transmitted on the kth ($k\in\mathcal{K}_D$) subcarrier as s_k, the VLC signal can be represented as

$$\begin{aligned} s^C(t) &= \sum_{k\in\mathcal{K}_D} s_k\exp(j2\pi k\Delta f_v t) + s_k^*\exp(-j2\pi k\Delta f_v t) \\ &= \sum_{k\in\mathcal{K}_D} 2\times\begin{Bmatrix}\mathrm{Re}\{s_k\}\cos(2\pi k\Delta f_v t)\\ -\mathrm{Im}\{s_k\}\sin(2\pi k\Delta f_v t)\end{Bmatrix}. \end{aligned} \tag{2}$$

Without loss of generality, we assume that $s_k\sim\mathcal{CN}(0,1)$. Then, according to the central limit theorem, $s^C(t)$ closely follows a real-valued zero-mean Gaussian distribution for the case of a large number of subcarriers [31], i.e., $s^C(t)\sim\mathcal{CN}(0,p_s)$, with $p_s=2|\mathcal{K}_D|$.

After the VLC signal is generated, a clipping operation will be performed on it, which is necessary to ensure that the OFDM modulated signal is nonnegative after DC bias. Following [32] and [33], the clipped VLC signal is given as

$$s_{\text{clip}}(t)=\begin{cases} s^C(t), & s^C(t)>-t_0\sqrt{p_s}\\ -t_0\sqrt{p_s}, & s^C(t)\le -t_0\sqrt{p_s}\end{cases} \tag{3}$$

where t_0 is called the normalized clipping level. Then, for the mth LED, the transmitted signal $s_m(t)$ is a linear combination of $s_{\text{clip}}(t)$ and the corresponding VLP signal $s_m^P(t)$, with a DC bias I_{DC} injected, i.e.,

$$s_m(t)=a_m s_{\text{clip}}(t)+b_m s_m^P(t)+I_{\text{DC}} \tag{4}$$

where $\mathbf{a}=[a_1,\ldots,a_M]^{\mathrm{T}}\in\mathbb{R}^M$ and $\mathbf{b}=[b_1,\ldots,b_M]^{\mathrm{T}}\in\mathbb{R}^M$ are the power allocation vectors for VLC and VLP signals, respectively. Moreover, to ensure the nonnegativity of $s_m(t)$, a_m and b_m should satisfy the following requirement:

$$t_0\sqrt{p_s}a_m+2b_m=t_0\sqrt{2|\mathcal{K}_D|}a_m+2b_m\le I_{\text{DC}}\quad\forall m. \tag{5}$$

Next, let us discuss the received signal model of the user. Since the VLP signal of each LED is transmitted through a dedicated subcarrier, the user can extract the received VLP signal corresponding to the mth LED [34], which is

$$y_m^p(t)=R_p h_m b_m s_m^p(t-\tau_m)+n_m(t) \tag{6}$$

where R_p is the responsivity of PD and $n_m(t)$ is zero-mean additive white Gaussian noise (AWGN) with a power spectral

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density (PSD) of σ_v^2. Besides, h_m denotes the optical channel gain from the mth LED to the user, i.e.,

$$h_m = \frac{(\varpi+1)S}{2\pi\left\|\boldsymbol{l}_r-\boldsymbol{l}_t^m\right\|^2}\cos^{\varpi}(\phi_m)\cos(\varphi_m) = \frac{(\varpi+1)S}{2\pi}\frac{\left[\left(\boldsymbol{l}_r-\boldsymbol{l}_t^m\right)^{\mathrm{T}}\mathbf{v}_t^m\right]^{\varpi}\left(\boldsymbol{l}_r-\boldsymbol{l}_t^m\right)^{\mathrm{T}}\mathbf{v}_r}{\left\|\boldsymbol{l}_r-\boldsymbol{l}_t^m\right\|^{(\varpi+3)}} \tag{7}$$

and τ_m is the time-delay of the mth VLP signal, which is calculated as

$$\tau_m = \frac{\left\|\boldsymbol{l}_r-\boldsymbol{l}_t^m\right\|}{c} + \Delta\tau_m. \tag{8}$$

In (7), ϖ is the order of the Lambertian emission and S is the detector area of the PD receiver. ϕ_m and φ_m represent the radiation angle and incident angle between the mth LED and the user, respectively. $\boldsymbol{l}_r = [l_{r,1}, l_{r,2}, H_r]^{\mathrm{T}}$ and $\boldsymbol{l}_t^m = [l_{t,1}^m, l_{t,2}^m, H_t]^{\mathrm{T}}$ are the coordinate vectors of the user and the mth LED, respectively, with H_r and H_t representing the heights of the PD receiver and the ceiling. $\mathbf{v}_r$ and $\mathbf{v}_t^m$ are the orientation vectors of the user and the mth LED, respectively [35]. In (8), $\Delta\tau_m$ is the time offset between the mth LED transmitter and the user.

The VLC signal received by the user at the kth ($k \in \mathcal{K}_D$) subcarrier is given as[1]

$$y_k^c = R_{pd}\left(\sum_{m=1}^{M} h_m a_m\right)s_{\mathrm{clip},k} + n_k, \quad k \in \mathcal{K}_D \tag{9}$$

where $s_{\mathrm{clip},k} = As_k + n_{\mathrm{clip},k}$ reflects the clipping process in the frequency domain [36]. The attenuation coefficient is $A = 1 - Q(t_0)$ and the clipping noise is $n_{\mathrm{clip},k} \sim \mathcal{CN}(0, p_{\mathrm{clip}})$, with

$$p_{\mathrm{clip}} = \left(1+t_0^2\right)Q(-t_0) - \left(1+t_0^2\right)Q^2(-t_0) - 2t_0\cdot\phi(t_0)\,Q(-t_0) + t_0\cdot\phi(t_0) - \phi^2(t_0) \tag{10}$$

where $\phi(\cdot)$ and $Q(\cdot)$ denote the probability density function and the tail probability of the standard Gaussian distribution.

B. Signal Model of the RF System

Based on the IEEE 802.11 standard protocols that have been widely promoted in the wireless local area networks, we consider that the indoor RF system adopts the OFDM technique for signal modulation. Assume that the RF link has Q subcarriers with unit bandwidth Δf_r. We use x_q and $\mathbf{v}_q \in \mathbb{C}^{N\times 1}$ to represent the transmitted symbol and beamformer corresponding to the qth subcarrier. Accordingly, the signal emitted by the RF transmitter is

$$\mathbf{x}(t) = \sum_{q=1}^{Q}\mathbf{v}_q x_q \exp\{j2\pi(f_c + q\Delta f_r)t\} \tag{11}$$

and the RF signal received on the qth subcarrier can be formulated as

$$y_q^{\mathrm{RF}} = \mathbf{g}_q^H\mathbf{v}_q x_q + n_q \tag{12}$$

[1]We consider the case of flat fading VLC channel, which is valid for most VLC systems that use the flat part, namely, 20 MHz [36]. Thus, for a specified LED, all the subcarriers have identical channel responses.

where $\mathbf{g}_q \in \mathbb{C}^{N\times 1}$ is the channel transfer function of the qth subcarrier. The noise n_q obeys a zero-mean Gaussian distribution, i.e., $n_q \sim \mathcal{CN}(0, \sigma_r^2\Delta f_r)$, where σ_r^2 denotes the PSD of the noise.

III. Problem Formulation

In this section, we first discuss the performance metrics regarding the positioning accuracy, the data rate and the EE of the hybrid RF-VLCP system. Then, an optimization problem is formulated to optimize or control them.

A. Positioning Accuracy

In this article, we adopt the RSS measurement for positioning. Compared to other characteristics of the received signals (i.e., TOA, TDOA, and AOA), the RSS value is easily obtained with a single PD and has no requirement on time synchronization. In addition, the RSS measurement is particularly suitable for indoor environment since there is a dominant line-of-sight (LOS) path between the LED and the user. As a result, high positioning accuracy can be expected by adopting the RSS measurement. Assuming that 1) the LEDs illuminate vertically downward and 2) the heights of VLC receiver and LEDs are known, the CRLB of the positioning error can be expressed as [35]

$$\mathbb{E}\left\{|\hat{l}_{r,1}-l_{r,1}|^2 + |\hat{l}_{r,2}-\hat{l}_{r,2}|^2\right\} \geq \mathrm{trace}\left\{J^{-1}(\mathbf{p}_b)\right\} \tag{13}$$

where

$$J(\mathbf{p}_b) = (\mathbf{I}_2\otimes\mathbf{p}_b)^{\mathrm{T}}\Gamma = (\mathbf{I}_2\otimes\mathbf{p}_b)^{\mathrm{T}}\begin{pmatrix}\Gamma_{11}, & \Gamma_{12}\\ \Gamma_{21}, & \Gamma_{22}\end{pmatrix} \tag{14}$$

and $\mathbf{p}_b = [b_1^2, \ldots, b_M^2]^{\mathrm{T}}$. Besides, $\Gamma_{ij} = [\Gamma_{ij}^{(1)}, \ldots, \Gamma_{ij}^{(M)}]^{\mathrm{T}}$ is an $M\times 1$ vector with $\Gamma_{ij}^{(m)}$ given as

$$\Gamma_{ij}^{(m)} = \frac{R_p^2}{\sigma_v^2}\left(E_{2,m} - \frac{\left(E_{3,m}\right)^2}{E_{1,m}}\right)\frac{\partial h_m}{\partial l_{r,i}}\frac{\partial h_m}{\partial l_{r,j}} \tag{15}$$

where

$$\frac{\partial h_m}{\partial l_{r,i}} = \frac{(\varpi+1)S}{2\pi}\times\left\{-\frac{(H_t-H_r)^{\varpi}v_{r,i}}{\left\|\boldsymbol{l}_r-\boldsymbol{l}_t^m\right\|^{(\varpi+3)}} + (\varpi+3) \times\frac{(H_t-H_r)^{\varpi}\left(\boldsymbol{l}_r-\boldsymbol{l}_t^m\right)^{\mathrm{T}}\mathbf{v}_r}{\left\|\boldsymbol{l}_r-\boldsymbol{l}_t^m\right\|^{(\varpi+5)}}\left(l_{r,i}-l_{t,i}^m\right)\right\} \tag{16a}$$

$$E_{1,m} = \int_{T_p}\left(\bar{s}_m^p(t)\right)^2 dt = 2T_p \tag{16b}$$

$$E_{2,m} = \int_{T_p}\left(s_m^p(t)\right)^2 dt = 2T_p \tag{16c}$$

$$E_{3,m} = \int_{T_p} s_m^p(t)\bar{s}_m^p(t)dt = 0 \tag{16d}$$

with $\bar{s}_m^p(t)$ denoting the derivative of $s_m^p(t)$ and T_p denoting the duration of the VLP signal received for one positioning. Based on (13), the CRLB constraint is formulated as

$$\mathrm{CRLB}_{\mathrm{RSS}} = \mathrm{trace}\left\{\left[(\mathbf{I}\otimes\mathbf{p}_b)^{\mathrm{T}}\Gamma\right]^{-1}\right\} \leq \varepsilon \tag{17}$$

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where ε represents the maximum tolerable CRLB of the positioning error.

Remark: To construct the matrix Γ, the central controller needs the information of several parameters, as shown in (15) and (16). Except for the user's location $\boldsymbol{l}_r$ and orientation $\mathbf{v}_r$, the other parameters can be assumed to be perfectly known by the central controller. In practice, a rough estimation of the user's location (denoted as $\hat{\boldsymbol{l}}_r$) will be used to substitute $\boldsymbol{l}_r$, which can be the result of last positioning or be obtained by GPS [37]. Moreover, the information of $\hat{\boldsymbol{l}}_r$ and $\mathbf{v}_r$ can be periodically transmitted from the user to the RF system via uplink communication, and then be further transmitted to the central controller.

On the other hand, since the RSS measurement is used for positioning, the receiver also needs the information of $\mathbf{p}_b$ to process the received VLP signal. Thus, the information is arranged to be transmitted via the VLC link at the same time as the VLP link transmits the VLP signal.

B. Data Rate and Energy Efficiency

According to (12) and (9), the data rates of the RF and VLC links are, respectively, calculated as

$$R_{\mathrm{RF}} = \Delta f_r \sum_{q=1}^{Q} \log_2\left(1 + \frac{\left|\mathbf{g}_q^H \mathbf{v}_q\right|^2}{\sigma_r^2 \Delta f_r}\right)$$
$$R_{\mathrm{VLC}} = |\mathcal{K}_D| \Delta f_v \log_2\left(1 + \frac{A^2 R_{pd}^2 \left(\mathbf{h}^{\mathrm{T}}\mathbf{a}\right)^2}{p_{\mathrm{clip}} R_{pd}^2 \left(\mathbf{h}^{\mathrm{T}}\mathbf{a}\right)^2 + \sigma_v^2 \Delta f_v}\right) \quad (18)$$

where $\mathbf{h} = [h_1, \ldots, h_M]^{\mathrm{T}}$. The transmit electrical power of each link is

$$P_{\mathrm{RF}} = \mu_r^{-1} \mathbb{E}\left\{|x(t)|^2\right\} = \mu_r^{-1} \sum_{q=1}^{Q} \left\|\mathbf{v}_q\right\|^2 \quad (19a)$$
$$P_{\mathrm{VLC}} = \mu_v^{-1} \|\mathbf{a}\|^2 \mathbb{E}\left\{|s_{\mathrm{clip}}(t)|^2\right\} \le 2\mu_v^{-1} |\mathcal{K}_D| \|\mathbf{a}\|^2 \quad (19b)$$
$$P_{\mathrm{VLP}} = \mu_v^{-1} \sum_m b_m^2 \mathbb{E}\left\{|s_m^P(t)|^2\right\} = 2\mu_v^{-1} \|\mathbf{b}\|^2 \quad (19c)$$

where μ_v and μ_r are the drain efficiencies of the power amplifiers for the LEDs and RF transmitter, respectively.

The EE of the hybrid RF-VLCP system is defined as the ratio of the total rate provided by the system to the total power consumed by the system, which is

$$EE_{hy} = \frac{R_{\mathrm{RF}} + R_{\mathrm{VLC}}}{P_{\mathrm{RF}} + P_{\mathrm{VLC}} + P_{\mathrm{VLP}} + P_c} \quad (20)$$

where $P_c = P_{c,\mathrm{LED}} + P_{c,\mathrm{RF}}$, with $P_{c,\mathrm{LED}}$ and $P_{c,\mathrm{RF}}$ being the circuit energy consumption of LEDs and RF device, respectively.

C. Problem Formulation

The main purpose of this work is to maximize the EE of the hybrid system while guaranteeing the data rate and positioning accuracy. To do so, we develop a joint optimization problem w.r.t the beamformer of RF transmitter ($\mathbf{v}_q$) and the power allocation vectors of VLCP transmitter ($\mathbf{a}$ and $\mathbf{b}$), which is

$$\max_{\{\mathbf{v}_q\},\mathbf{a},\mathbf{b}} \quad \frac{R_{\mathrm{RF}} + R_{\mathrm{VLC}}}{P_{\mathrm{RF}} + P_{\mathrm{VLC}} + P_{\mathrm{VLP}} + P_c} \quad (21a)$$
$$\text{s.t.} \quad \mathrm{CRLB}_{\mathrm{RSS}} \le \varepsilon, \quad R_{\mathrm{RF}} + R_{\mathrm{VLC}} \ge R_{\min} \quad (21b)$$
$$\mu_r^{-1} \sum_{q=1}^{Q} \left\|\mathbf{v}_q\right\|^2 \le P_{\mathrm{RF,max}} \quad (21c)$$
$$2\mu_v^{-1} |\mathcal{K}_D| \|\mathbf{a}\|^2 + 2\mu_v^{-1} \|\mathbf{b}\|^2 \le P_{\mathrm{LED,max}} \quad (21d)$$
$$t_0 \sqrt{2|\mathcal{K}_D|} a_m + 2b_m \le I_{\mathrm{DC}} \quad \forall m \quad (21e)$$
$$a_m \ge 0, \quad b_m \ge 0 \quad \forall m \quad (21f)$$

where $P_{\mathrm{RF,max}}$ and $P_{\mathrm{LED,max}}$ denote the maximum allowable transmission powers for the RF transmitter and the LEDs, respectively. Constraint (21e) ensures the nonnegativity of the signal transmitted by each LED.

Remark: Problem (21) is a nonconvex problem and challenging to solve. However, we denote that the DSAC algorithm proposed in [29] can be extended as a potential solution in a special case. When t_0 is relatively large, the effect of the clipping operation on the frequency domain signal can be neglected [33], meaning that $s_{\mathrm{clip},k}$ in (9) is almost the same to s_k. As a result, the data rate R_{VLC} in (18) can be well approximated by $R_{\mathrm{VLC}}^{\mathrm{appr}} = |K_D| \Delta f_v \log_2(1 + R_{pd}^2 (\mathbf{h}^{\mathrm{T}}\mathbf{a})^2 / (\sigma_v^2 \Delta f_v))$. If $R_{\mathrm{VLC}}^{\mathrm{appr}}$ is used in (21a) and (21b), problem (21) can be iteratively solved by the DSAC algorithm, and a stationary point of the problem can be found. Briefly speaking, the DSAC algorithm solves the following problem in the $(i+1)$th iteration

$$\max_{\{\mathbf{v}_q\},\mathbf{a},\mathbf{b}} \quad \left(R_{\mathrm{RF}} + R_{\mathrm{VLC}}^{\mathrm{appr}}\right) - e^{(i)} (P_{\mathrm{RF}} + P_{\mathrm{VLC}} + P_{\mathrm{VLP}} + P_c)$$
$$\text{s.t.} \quad \mathrm{CRLB}_{\mathrm{RSS}} \le \varepsilon, \quad R_{\mathrm{RF}} + R_{\mathrm{VLC}}^{\mathrm{appr}} \ge R_{\min}$$
$$(21c) - (21f) \quad (22)$$

where $e^{(i)} = (R_{\mathrm{RF}}^{(i)} + R_{\mathrm{VLC}}^{\mathrm{appr}\,(i)}) / (P_{\mathrm{RF}}^{(i)} + P_{\mathrm{VLC}}^{(i)} + P_{\mathrm{VLP}}^{(i)} + P_c)$ is obtained based on the result of the ith iteration $(\mathbf{v}_q^{(i)}, \mathbf{a}^{(i)}, \mathbf{b}^{(i)})$. Since problem (22) is still nonconvex, it is then approximated as a convex problem (23), shown at the bottom of the next page, which can be solved with a standard convex solver, i.e., CVX.

However, the DSAC algorithm suffers from the following two deficiencies.

1) Solving problem (23), with an existing mature solver may consume a significantly long time, which is highly undesirable in the considered scenario. Since an estimate of user's coordinate ($\hat{\boldsymbol{l}}_r$) is employed to characterize the CRLB metric in (21b), the VLP signal is expected to be sent immediately after $\hat{\boldsymbol{l}}_r$ is obtained. A long processing time for the problem (21) may further enhance the inaccuracy of $\hat{\boldsymbol{l}}_r$ because the user may move around during this time. As a result, the positioning performance of the VLP signal will seriously degrade.
2) For small t_0 case, R_{VLC} cannot be substituted by $R_{\mathrm{VLC}}^{\mathrm{appr}}$, and accordingly, problem (22) cannot be approximated by a convex problem. Therefore, the DSAC algorithm cannot be used for a small t_0 case.

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Based on the above reasons, we next introduce a novel algorithm that can tackle problem (21) more quickly. Besides, the proposed algorithm dose not make an approximation to R_{VLC} and is therefore applicable for any value of t_0.

IV. Our Proposed Algorithm

The proposed algorithm is a two-step method. The first step determines the power allocation of the VLP signal, followed by the second step to optimize the power allocation of VLC signal and the beamformer of RF signal.

A. First Step Is to Determine $\boldsymbol{b}$

As can be seen from (21), the power optimization of the VLP and VLC signals is coupled due to the constraints (21d) and (21e). Considering that the positioning accuracy is guaranteed solely by the VLP signal and the data rate can be guaranteed jointly by the VLP and RF signals, it is reasonable to prioritize the power resources of the LEDs to the VLP signal. To this end, we first optimize $\mathbf{b}$ by solving the following problem:

$$\begin{aligned} \min_{\mathbf{p}_b} \quad & \mathbf{1}^{\mathrm{T}}\mathbf{p}_b \\ \text{s.t.} \quad & \mathrm{trace}\left\{\left[\left(\mathbf{I}\otimes\mathbf{p}_b\right)^{\mathrm{T}}\Gamma\right]^{-1}\right\} \le \varepsilon \\ & 0 \le \mathbf{p}_b[m] \le \Lambda \quad \forall m \end{aligned} \tag{24}$$

where $\Lambda = I_{\mathrm{DC}}^2/4$. Problem (24) is a convex problem, thus it can be sufficiently solved via CVX solver. However, as we have mentioned before, the calculation speed of the algorithm is very important for considered scenario. Therefore, in the following, we introduce an ADMM-based algorithm that can be much faster than the CVX solver and is promised to find the globally optimal solution of problem (24) with a mild condition. The principle of ADMM algorithm can be found in [38].

To begin with, we reformulate the CRLB metric as

$$\begin{aligned} \mathrm{CRLB}_{\mathrm{RSS}} &= \frac{2\sum_{m=1}^{M}\mathbf{p}_b[m]\left(\left(\chi_1^m\right)^2+\left(\chi_2^m\right)^2\right)}{\sum_{m=1}^{M}\sum_{n=1}^{M}\mathbf{p}_b[n]\mathbf{p}_b[m]\left(\chi_1^n\chi_2^m-\chi_2^n\chi_1^m\right)^2} \\ &= \frac{\boldsymbol{\gamma}^{\mathrm{T}}\mathbf{p}_b}{\mathbf{p}_b^{\mathrm{T}}\boldsymbol{\Upsilon}\mathbf{p}_b} \end{aligned} \tag{25}$$

where $\chi_i^m = (R_p/\sigma_v)\sqrt{E_{2,m}-(E_{3,m}^2/E_{1,m})}([\partial h_m]/[\partial l_{r,i}])$, $\boldsymbol{\Upsilon}[m,n] = ([(\chi_1^n\chi_2^m-\chi_2^n\chi_1^m)^2]/2)$ and $\boldsymbol{\gamma}[m] = (\chi_1^m)^2 + (\chi_2^m)^2$. The detailed derivation of (25) is given in Appendix A.

Based on (25), we can equivalently convert problem (24) to

$$\begin{aligned} \min_{\mathbf{p}_b,\mathbf{z}_1,\mathbf{z}_2} \quad & \mathbf{1}^{\mathrm{T}}\mathbf{p}_b \\ \text{s.t.} \quad & -\mathbf{z}_1^{\mathrm{T}}\varepsilon\boldsymbol{\Upsilon}\mathbf{z}_1+\boldsymbol{\gamma}^{\mathrm{T}}\mathbf{z}_1 \le 0 \\ & 0 \le \mathbf{z}_2[m] \le \Lambda \quad \forall m \\ & \mathbf{z}_q = \mathbf{p}_b, \quad q=1,2. \end{aligned} \tag{26}$$

The augmented Lagrangian function of problem (26) is

$$\mathcal{L}_A\left(\mathbf{p}_b,\{\mathbf{z}_q\},\{\boldsymbol{\mu}_q\}\right) = \mathbf{1}^{\mathrm{T}}\mathbf{p}_b + \rho\sum_{q=1}^{2}\left\|\mathbf{z}_q-\mathbf{p}_b+\boldsymbol{\mu}_q\right\|^2 \tag{27}$$

where ρ is a penalty parameter. Then, the process of the $(n+1)$th ADMM iteration is given as follows:

$$\mathbf{p}_b^{(n+1)} := \arg\min_{\mathbf{p}_b} \ \mathcal{L}_A\left(\mathbf{p}_b,\mathbf{z}_1^{(n)},\mathbf{z}_2^{(n)},\left\{\boldsymbol{\mu}_q^{(n)}\right\}\right) \tag{28a}$$

$$\begin{aligned} \mathbf{z}_1^{(n+1)} := \arg\min_{\mathbf{z}_1} \ & \mathcal{L}_A\left(\mathbf{p}_b^{(n+1)},\mathbf{z}_1,\mathbf{z}_2^{(n)},\left\{\boldsymbol{\mu}_q^{(n)}\right\}\right) \\ \text{s.t.} \quad & -\mathbf{z}_1^{\mathrm{T}}\varepsilon\boldsymbol{\Upsilon}\mathbf{z}_1+\gamma^{\mathrm{T}}\mathbf{z}_1 \le 0 \end{aligned} \tag{28b}$$

$$\begin{aligned} \mathbf{z}_2^{(n+1)} := \arg\min_{\mathbf{z}_2} \ & \mathcal{L}_A\left(\mathbf{p}_b^{(n+1)},\mathbf{z}_1^{(n+1)},\mathbf{z}_2,\left\{\boldsymbol{\mu}_q^{(n)}\right\}\right) \\ \text{s.t.} \quad & 0 \le \mathbf{z}_2[m] \le \Lambda \quad \forall m \end{aligned} \tag{28c}$$

$$\boldsymbol{\mu}_q^{(n+1)} := \boldsymbol{\mu}_q^{(n)} + \mathbf{z}_q^{(n+1)} - \mathbf{p}_b^{(n+1)}, \quad q=1,2. \tag{28d}$$

In the following, we will discuss the solutions of subproblems (28a)–(28c) in detail.

1) Solution of (28a): Letting $\partial\mathcal{L}_A(\mathbf{p}_b,\mathbf{z}_1^{(n)},\mathbf{z}_2^{(n)},\{\boldsymbol{\mu}_q^{(n)}\})/\partial\mathbf{p}_b = \mathbf{0}$, we have

$$\mathbf{p}_b^{(n+1)} = -\frac{\mathbf{1}}{4\rho}+\frac{1}{2}\sum_{q=1}^{2}\left(\mathbf{z}_q^{(n)}+\boldsymbol{\mu}_q^{(n)}\right). \tag{29}$$

2) Solution of (28b): First, problem (28b) is equivalent to the following problem:

$$\min_{\mathbf{z}_1} \quad \|\mathbf{z}_1-\mathbf{p}_b^{(n+1)}+\boldsymbol{\mu}_1^{(n)}\|^2 \tag{30a}$$

$$\text{s.t.} \quad -\mathbf{z}_1^{\mathrm{T}}\varepsilon\Upsilon\mathbf{z}_1+\boldsymbol{\gamma}^{\mathrm{T}}\mathbf{z}_1 \le 0. \tag{30b}$$

Denoting $\boldsymbol{\eta}^{(n)} = \mathbf{p}_b^{(n+1)}-\boldsymbol{\mu}_1^{(n)}$, if $\boldsymbol{\eta}^{(n)}$ satisfies constraint (30b), then it is the optimal solution of (30a). In this case, $\mathbf{z}_1$

$$\begin{aligned} \max_{\{\mathbf{v}_q\},\,\mathbf{a},\,\mathbf{b},\,\{\bar{R}_{\mathrm{RF},q}\},\,\bar{R}_{\mathrm{VLC}}^{\mathrm{appr}},\,\{\phi_q\},\,\varsigma,\,\mathbf{p}_b} \quad & \sum_q \bar{R}_{\mathrm{RF},q}+\bar{R}_{\mathrm{VLC}}^{\mathrm{appr}}-e^{(i)}(P_{\mathrm{RF}}+P_{\mathrm{VLC}}+P_{\mathrm{VLP}}+P_c) \\ \text{s.t.} \quad & \sum_q \bar{R}_{\mathrm{RF},q}+\bar{R}_{\mathrm{VLC}}^{\mathrm{appr}} \ge R_{\min} \\ & \bar{R}_{\mathrm{RF},q} \le \Delta f_r\log_2\left(1+\frac{\phi_q}{\sigma_r^2\Delta f_r}\right), \quad \phi_q \le \mathbf{v}_q^{(i)H}\mathbf{g}_q\mathbf{g}_q^H\mathbf{v}_q^{(i)}+2\mathrm{Re}\left\{\mathbf{v}_q^{(i)H}\mathbf{g}_q\mathbf{g}_q^H\left(\mathbf{v}_q-\mathbf{v}_q^{(i)}\right)\right\} \quad \forall q \\ & \bar{R}_{\mathrm{VLC}}^{\mathrm{appr}} \le |\mathcal{K}_D|\Delta f_v\log_2\left(1+\frac{A^2R_{pd}^2\,\varsigma}{\sigma_v^2\Delta f_v}\right), \quad \varsigma \le \mathbf{a}^{(i)T}\mathbf{h}\mathbf{h}^{\mathrm{T}}\mathbf{a}^{(i)}+2\mathbf{a}^{(i)T}\mathbf{h}\mathbf{h}^{\mathrm{T}}\left(\mathbf{a}-\mathbf{a}^{(i)}\right) \\ & \mathrm{trace}\left\{\left[\left(\mathbf{I}\otimes\mathbf{p}_b\right)^{\mathrm{T}}\Gamma\right]^{-1}\right\} \le \varepsilon, \quad p_{b,m} \le b_m^{(i)2}+2b_m^{(i)}\left(b_m-b_m^{(i)}\right) \quad (21\mathrm{d})-(21\mathrm{f}) \end{aligned} \tag{23}$$

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is updated as $\mathbf{z}_1^{(n+1)} = \boldsymbol{\eta}^{(n)}$. If $\boldsymbol{\eta}^{(n)}$ does not satisfy constraint (30b), then the constraint must be satisfied as equality [39]. Therefore, we can resort to the following optimization problem:

$$\mathbf{z}_1^{(n+1)} := \arg \min_{\mathbf{z}_1} \quad \|\mathbf{z}_1 - \boldsymbol{\eta}^{(n)}\|^2 \tag{31a}$$

$$\text{s.t.} \quad -\mathbf{z}_1^{\mathrm{T}} \varepsilon \Upsilon \mathbf{z}_1 + \boldsymbol{\gamma}^{\mathrm{T}} \mathbf{z}_1 = 0. \tag{31b}$$

Notice that $\varepsilon\Upsilon$ is a real symmetric matrix, so it can be decomposed as $\varepsilon\Upsilon = \mathbf{Q}^{\mathrm{T}}\Xi\mathbf{Q}$, where Ξ is a real diagonal matrix and $\mathbf{Q}$ is an orthogonal matrix. Then, defining $\tilde{\mathbf{z}}_1 = \mathbf{Q}^{\mathrm{T}}\mathbf{z}_1$, $\tilde{\boldsymbol{\eta}}^{(n)} = \mathbf{Q}^{\mathrm{T}}\boldsymbol{\eta}^{(n)}$, and $\tilde{\boldsymbol{\gamma}} = \mathbf{Q}^{\mathrm{T}}\boldsymbol{\gamma}$, the Lagrangian function of problem (31) is given as

$$\mathcal{L}_1^{(n)}(\tilde{\mathbf{z}}_1, \nu) = \|\tilde{\mathbf{z}}_1 - \tilde{\boldsymbol{\eta}}^{(n)}\|^2 + \nu\left(-\tilde{\mathbf{z}}_1^{\mathrm{T}}\Xi\tilde{\mathbf{z}}_1 + \tilde{\boldsymbol{\gamma}}^{\mathrm{T}}\tilde{\mathbf{z}}_1\right). \tag{32}$$

Letting $\partial\mathcal{L}_1^{(n)}/\partial\tilde{\mathbf{z}}_1 = \mathbf{0}$, we have

$$\tilde{\mathbf{z}}_1^{(n+1)} = (\mathbf{I} - \nu\Xi)^{-1}\left(\tilde{\boldsymbol{\eta}}^{(n)} - \nu\tilde{\boldsymbol{\gamma}}\right). \tag{33}$$

Then, substituting (33) into (31b), it can be derived that ν should satisfy

$$\sum_{m=1}^{M} -\lambda_m \frac{\left(\tilde{\eta}_m^{(n)} - \nu\tilde{\gamma}_m\right)^2}{(1-\nu\lambda_m)^2} + \tilde{\gamma}_m \frac{\tilde{\eta}_m^{(n)} - \nu\tilde{\gamma}_m}{1-\nu\lambda_m} = 0 \tag{34}$$

where λ_m denotes the (m,m)th element of Ξ. $\tilde{\eta}_m^{(n)}$ and $\tilde{\gamma}_m$ represent the mth element of $\tilde{\boldsymbol{\eta}}^{(n)}$ and $\tilde{\boldsymbol{\gamma}}$, respectively. Problem (34) can be easily solved with bi-section method. After obtaining ν, one can calculate $\tilde{\mathbf{z}}_1^{(n+1)}$ based on (33) and then update $\mathbf{z}_1^{(n+1)} = \mathbf{Q}\tilde{\mathbf{z}}_1^{(n+1)}$.

3) Solution of (28c): Now, we need to solve the following problem:

$$\min_{\mathbf{z}_2} \quad \|\mathbf{z}_2 - \mathbf{p}_b^{(n+1)} + \boldsymbol{\mu}_2^{(n)}\|^2 \tag{35a}$$

$$\text{s.t.} \quad 0 \le \mathbf{z}_2[m] \le \Lambda \quad \forall\, m. \tag{35b}$$

Denoting $\boldsymbol{\varsigma}^{(n)} = \mathbf{p}_b^{(n+1)} - \boldsymbol{\mu}_2^{(n)}$, the optimal solution of (35) is given as follows:

$$\mathbf{z}_2[m] = \begin{cases} \varsigma^{(n)}[m], & \text{if } 0 \le \varsigma^{(n)}[m] \le \Lambda \\ 0, & \text{if } \varsigma^{(n)}[m] < 0 \\ \Lambda, & \text{if } \varsigma^{(n)}[m] > \Lambda. \end{cases} \tag{36}$$

Finally, the above process for determining $\mathbf{b}$ is concluded as Algorithm 1. And the convergence criterion for Algorithm 1 is given as follows [38]:

$$\frac{1}{M}\|\mathbf{p}_b^{(n+1)} - \mathbf{p}_b^{(n)}\|^2 \le \xi_{\text{prim}}^2 \tag{37a}$$

$$\frac{1}{M}\|\mathbf{z}_q^{(n+1)} - \mathbf{p}_b^{(n+1)}\|^2 \le \xi_{\text{dual}}^2 \quad \forall q \tag{37b}$$

where (37a) and (37b) are regarding to the primal and dual residuals, respectively.

Remark: It has been theoretically verified that the ADMM iteration is able to find the globally optimal solution for a convex problem, provided it eventually converges. And the convergence of ADMM iteration is closely related to the setting of penalty parameter ρ. If ρ is too small, the ADMM iteration may fail to converge. On the other hand, a bigger ρ is able to prevent the iteration from diverging toward infeasibility, but may take a long time to converge [40]. Thus, ρ should be set appropriately large. In addition, adjusting the penalty factor adaptively during iteration can be considered to promote convergence [41].

Algorithm 1 Obtaining $\mathbf{b}^\star$ Through ADMM-Based Algorithm

Input: thresholds ε and $\Lambda = I_{\text{DC}}^2/4$.

Initialize: $\{\mathbf{z}_q^{(0)}\} = \Lambda \cdot \mathbf{1}$, $\{\boldsymbol{\mu}_q^{(0)}\} = \mathbf{0}$, $n = 0$.

Calculate Υ and $\boldsymbol{\gamma}$.
while not converge, **do**
  update $\mathbf{p}_b^{(n+1)}$ with (29).
  update $\mathbf{z}_1^{(n+1)}$ by solving problem (30).
  update $\mathbf{z}_2^{(n+1)}$ with (36).
  update $\{\boldsymbol{\mu}_q^{(n+1)}\}$ with (28d).
  $n = n + 1$.
end while

Output: $b_m^\star = \sqrt{\mathbf{p}_b^{(n)}[m]}$, $\forall m$.

B. Second Step Is to Determine $\{\mathbf{v}_q\}$ and $\mathbf{a}$

After determining $\mathbf{b}^\star$, the electrical power consumed by VLP is given by $P_{\text{VLC}}^\star = 2\mu_v^{-1}\|\mathbf{b}^\star\|^2$. Next, we design $\{\mathbf{v}_q\}$ and $\mathbf{a}$ by resolving the following optimization problem:

$$\max_{\{\mathbf{v}_q\},\mathbf{a}} \quad \frac{R_{\text{RF}} + R_{\text{VLC}}}{P_{\text{RF}} + P_{\text{VLC}} + P_{\text{VLP}}^\star + P_c} \tag{38a}$$

$$\text{s.t.} \quad R_{\text{RF}} + R_{\text{VLC}} \ge R_{\min} \tag{38b}$$

$$\mu_r^{-1}\sum_q \|\mathbf{v}_q\|^2 \le P_{\text{RF,max}} \tag{38c}$$

$$t_0\sqrt{2|\mathcal{K}_D|}a_m \le I_{DC} - 2b_m^\star \quad \forall\, m \tag{38d}$$

$$2\mu_v^{-1}|\mathcal{K}_D|\|\mathbf{a}\|^2 \le P_{\text{LED,max}} - P_{\text{VLP}}^\star. \tag{38e}$$

According to [42], the optimal $\mathbf{v}_q^\star$ for the above problem should be formed as

$$\mathbf{v}_q^\star = \sqrt{p_q}\mathbf{g}_q/\|\mathbf{g}_q\|. \tag{39}$$

Then, to obtain the optimal solution of problem (38), we can resort to problem (40), shown at the bottom of the next page, where $\tilde{\mathbf{g}}_q = \frac{\mathbf{g}_q}{\sqrt{\sigma_r^2\Delta f_r}}$, $\beta = \frac{\Delta f_v}{\Delta f_r}|\mathcal{K}_D|$, $\tilde{R}_{\min} = \frac{R_{\min}}{\Delta f_r}$, $\phi_r = \mu_r^{-1}$, $\phi_v = 2\mu_v^{-1}|\mathcal{K}_D|$, and $\tau_m = \frac{(I_{DC}-2b_m^\star)^2}{2t_0^2|\mathcal{K}_D|}$, which optimizes the variables $\{p_q\}$ and $\{u_m = a_m^2\}$. A Dinkelbach-based algorithm for solving problem (40) is outlined in Algorithm 2, which finds the optimal solution of problem (40) through rough search and fine search. During the rough search, the data rate constraint (40b) is ignored, and the variables $\{p_q\}, \{u_m\}$ are updated within the framework of the Dinkelbach-based algorithm. This update process will make the rate value $\tilde{R}_{hy}$ in (40b) gradually decrease from its maximum value. When $\tilde{R}_{hy}$ becomes smaller than the threshold $\tilde{R}_{\min}$, the rough search ends. Then, the algorithm moves to the fine search, which is supposed to find a solution $\{p_q^\star\}, \{u_m^\star\}$ such that the corresponding $\tilde{R}_{hy}^\star$ is equal to $\tilde{R}_{\min}$. In Appendix B, we prove that

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Algorithm 2 Resolving (40) With Dinkelbach-Based Algorithm

Initialize: $\kappa^{(0)} = 0$, $i = 0$.
Rough search:

1: Update $\{p_q^{(i+1)}\}$, $\{u_m^{(i+1)}\}$ as the optimal solution of $\mathcal{P}_{\kappa^{(i)}}$.

$$\mathcal{P}_{\kappa^{(i)}} \begin{cases} \max_{\{p_q\},\{u_m\}} & \tilde{R}_{hy} - \kappa^{(i)} P_{hy} \\ \text{s.t.} & (40c) \end{cases}$$

2: Calculate $\tilde{R}_{hy}^{(i+1)}$ and $P_{hy}^{(i+1)}$ based on the solution of $\mathcal{P}_{\kappa^{(i)}}$, and update $\kappa^{(i+1)} = \tilde{R}_{hy}^{(i+1)}/P_{hy}^{(i+1)}$.

3: If $\tilde{R}_{hy}^{(i+1)} < \tilde{R}_{\min}$, end the rough search and go to the fine search. Otherwise, set $i{=}i{+}1$ and go back to step 1.

Fine search: set $\kappa_l{=}\kappa^{(i+1)}$, $\kappa_u{=}\kappa^{(i)}$ and then find a $\kappa^\star{\in}(\kappa_l, \kappa_u)$ such that the optimal solution of $\mathcal{P}_{\kappa^\star}$ (denoted as $\{p_q^\star\}$, $\{u_m^\star\}$) satisfies $\tilde{R}_{hy}^\star = \tilde{R}_{\min}$. There, the bisection method can be used to find $\kappa^\star$.
Output: $\{p_q^\star\}$, $\{u_m^\star\}$.

$\{p_q^\star\}$, $\{u_m^\star\}$ obtained by Algorithm 2 is the optimal solution of problem (40).

However, to use Algorithm 2, we still need to discuss the solution of $\mathcal{P}_{\kappa^{(i)}}$. For ease of expression, the superscript i will be omitted in the subsequent. We denote that $\mathcal{P}_\kappa$ is separable for $\{p_q\}$ and $\{u_m\}$, and the subproblem corresponding to $\{p_q\}$ is given as

$$\begin{aligned} \max_{\{p_q\}} \ & \sum_q \log_2\left(1 + p_q \|\tilde{\mathbf{g}}_q\|^2\right) - \kappa\, \phi_r \sum_q p_q \\ \text{s.t.} \ & \phi_r \sum_q p_q \le P_{\text{RF,max}}, \quad p_q \ge 0 \quad \forall q. \end{aligned} \tag{41}$$

The Lagrangian function of problem (41) is

$$\begin{aligned} \mathcal{L}_2(\{p_q\}, \nu_2, \{\tilde{\nu}_{2,q}\}) &= \sum_q \log_2\left(1 + p_q \|\tilde{\mathbf{g}}_q\|^2\right) - \kappa\phi_r \\ &\times \sum_q p_q - \nu_2\left(\phi_r \sum_q p_q - P_{\text{RF,max}}\right) + \sum_q \tilde{\nu}_{2,q} p_q. \end{aligned} \tag{42}$$

By letting $\partial\mathcal{L}_2/\partial p_q = 0$, the optimal $p_q^\star$ is given as

$$p_q^\star = \frac{1}{\ln 2 \cdot \left(\kappa\phi_r + \nu_2\phi_r - \tilde{\nu}_{2,q}\right)} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2}. \tag{43}$$

Then, depending on whether the constraints are active, we divide the solution (43) into four cases.

Case 1: $\nu_2 = 0$, $\tilde{\nu}_{2,q} = 0$ $\forall q$, which means that

$$\phi_r \sum_q p_q^\star < P_{\text{RF,max}}, \quad p_q^\star > 0 \quad \forall q. \tag{44}$$

For this case, the optimal $p_q^\star$ in (43) becomes

$$p_q^\star = \frac{1}{\ln 2 \cdot \kappa\phi_r} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2} \quad \forall q. \tag{45}$$

And combining (44) and (45), we can derive that this case happens when (46) is satisfied

$$\frac{Q}{\ln 2 \cdot \kappa} - \phi_r \sum_q \frac{1}{\|\tilde{\mathbf{g}}_q\|^2} < P_{\text{RF,max}} \tag{46a}$$

$$\|\tilde{\mathbf{g}}_q\|^2 > \ln 2 \cdot \kappa\phi_r \quad \forall q. \tag{46b}$$

Case 2: $\nu_2 \ne 0$, $\tilde{\nu}_{2,q} = 0$ $\forall q$, which means

$$\phi_r \sum_q p_q^\star = P_{\text{RF,max}}, \quad p_q^\star > 0 \quad \forall q. \tag{47}$$

For this case, the optimal $p_q^\star$ in (43) becomes

$$p_q^\star = \frac{1}{\ln 2 \cdot (\kappa\phi_r + \nu_2\phi_r)} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2} \quad \forall q. \tag{48}$$

And ν_2 is given as

$$\nu_2 = \frac{Q}{\ln 2\left(P_{\text{RF,max}} + \phi_r \sum_q 1/\|\tilde{\mathbf{g}}_q\|^2\right)} - \kappa. \tag{49}$$

Besides, we can derive that this case happens when (50) is satisfied

$$\frac{Q}{\ln 2 \cdot \kappa} - \phi_r \sum_q \frac{1}{\|\tilde{\mathbf{g}}_q\|^2} > P_{\text{RF,max}} \tag{50a}$$

$$\|\tilde{\mathbf{g}}_q\|^2 > \ln 2 \cdot \kappa\phi_r \quad \forall q. \tag{50b}$$

Case 3: $\nu_2 = 0$, $\tilde{\nu}_{2,q} = 0$, if $q \in \Theta$, $\tilde{\nu}_{2,q} > 0$, if $q \notin \Theta$, which means that

$$\phi_r \sum_q p_q^\star < P_{\text{RF,max}} \tag{51a}$$

$$p_q^\star > 0, \text{ if } q \in \Theta p_q^\star = 0, \text{ if } q \notin \Theta. \tag{51b}$$

$$\max_{\{p_q\},\{u_m = a_m^2\}} \frac{\tilde{R}_{hy}}{\tilde{P}_{hy}} = \frac{\sum_{q=1}^Q \log_2\left(1 + p_q\|\tilde{\mathbf{g}}_q\|^2\right) + \beta \log_2\left(1 + \frac{A^2 R_{pd}^2 \left(\sum_{m=1}^M h_m\sqrt{u_m}\right)^2}{p_{\text{clip}} R_{pd}^2 \left(\sum_{m=1}^M h_m\sqrt{u_m}\right)^2 + \sigma_v^2 \Delta f_v}\right)}{\phi_r \sum_{q=1}^Q p_q + \phi_v \sum_{m=1}^M u_m + P_{\text{VLP}}^\star + P_c} \tag{40a}$$

$$\text{s.t. } \tilde{R}_{hy} = \sum_{q=1}^Q \log_2\left(1 + p_q\|\tilde{\mathbf{g}}_q\|^2\right) + \beta \log_2\left(1 + \frac{A^2 R_{pd}^2 \left(\sum_{m=1}^M h_m\sqrt{u_m}\right)^2}{p_{\text{clip}} R_{pd}^2 \left(\sum_{m=1}^M h_m\sqrt{u_m}\right)^2 + \sigma_v^2 \Delta f_v}\right) \ge \tilde{R}_{\min} \tag{40b}$$

$$\phi_r \sum_{q=1}^Q p_q \le P_{\text{RF,max}}, \quad p_q \ge 0, \quad \phi_v \sum_{m=1}^M u_m \le P_{\text{LED,max}} - P_{\text{VLP}}^\star, \quad 0 \le u_m \le \tau_m \tag{40c}$$

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For $p_q^\star = 0$ and $\tilde{\nu}_{2,q} > 0$, we have

$$0 = p_q^\star > \frac{1}{\ln 2 \cdot (\kappa \phi_r)} - \frac{1}{\left\|\tilde{\mathbf{g}}_q\right\|^2}. \tag{52}$$

For $p_q^\star > 0$ and $\tilde{\nu}_{2,q} = 0$, we have

$$p_q^\star = \frac{1}{\ln 2 \cdot \kappa \phi_r} - \frac{1}{\left\|\tilde{\mathbf{g}}_q\right\|^2} > 0. \tag{53}$$

According to (52) and (53), it can be concluded that the optimal $p_q^\star$ in this case is

$$p_q^\star = \begin{cases} 0, & \text{if } \|\tilde{\mathbf{g}}_q\|^2 < \ln 2 \cdot \kappa \phi_r \\ \frac{1}{\ln 2 \cdot \kappa \phi_r} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2}, & \text{if } \|\tilde{\mathbf{g}}_q\|^2 > \ln 2 \cdot \kappa \phi_r. \end{cases} \tag{54}$$

Accordingly, Θ is given as $\Theta = \{q \mid \ln 2 \cdot \kappa \phi_r < \|\tilde{\mathbf{g}}_q\|^2\}$. And combining with (51a), it can be concluded that case 3 happens when (55) is satisfied

$$\frac{|\Theta|}{\ln 2 \cdot \kappa} - \phi_r \sum_{q \in \Theta} \frac{1}{\|\tilde{\mathbf{g}}_q\|^2} < P_{\mathrm{RF,max}}. \tag{55}$$

Case 4: $\nu_2 > 0$, $\tilde{\nu}_{2,q} = 0$, if $q \in \bar{\Theta}$, $\tilde{\nu}_{2,q} > 0$, if $q \notin \bar{\Theta}$, which means that

$$\phi_r \sum_q p_q^\star = P_{\mathrm{RF,max}} \tag{56a}$$

$$p_q^\star > 0, \text{ if } q \in \bar{\Theta}, \quad p_q^\star = 0, \text{ if } q \notin \bar{\Theta}. \tag{56b}$$

For $p_q^\star = 0$ and $\tilde{\nu}_{2,q} > 0$, we have

$$0 = p_q^\star > \frac{1}{\ln 2 \cdot (\kappa \phi_r + \nu_2 \phi_r)} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2}. \tag{57}$$

For $p_q^\star > 0$ and $\tilde{\nu}_{2,q} = 0$, we have

$$p_q^\star = \frac{1}{\ln 2 \cdot (\kappa \phi_r + \nu_2 \phi_r)} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2} > 0. \tag{58}$$

According to (57) and (58), it can be concluded that the optimal $p_q^\star$ in this case is

$$p_q^\star = \begin{cases} 0, & \text{if } \left\|\tilde{\mathbf{g}}_q\right\|^2 < \ln 2 \cdot \phi_r(\kappa + \nu_2) \\ \frac{1}{\ln 2 \cdot \phi_r(\kappa+\nu_2)} - \frac{1}{\|\tilde{\mathbf{g}}_q\|^2}, & \text{if } \left\|\tilde{\mathbf{g}}_q\right\|^2 > \ln 2 \cdot \phi_r(\kappa + \nu_2). \end{cases} \tag{59}$$

Moreover, given that $\phi_r \sum_q p_q = P_{\mathrm{RF,max}}$, ν_2 and $\bar{\Theta}$ are determined as follows. First, we define $\|\tilde{\mathbf{g}}_k'\|^2$ as the descending order of $\|\tilde{\mathbf{g}}_q\|^2$. Then, assuming that $\bar{\Theta}$ has $\bar{K}$ elements, we have

$$\ln 2 \cdot \phi_r(\kappa + \nu_2) = \frac{\bar{K}}{P_{\mathrm{RF,max}}/\phi_r + \sum_{k=1}^{\bar{K}} 1/\left\|\tilde{\mathbf{g}}_k'\right\|^2}. \tag{60}$$

Combining (59) and (60), $\bar{K}$ can be found based on the following condition:

$$\left\|\tilde{\mathbf{g}}_{\bar{K}+1}'\right\|^2 < \frac{\bar{K}}{P_{\mathrm{RF,max}}/\phi_r + \sum_{k=1}^{\bar{K}} 1/\left\|\tilde{\mathbf{g}}_k'\right\|^2}. \tag{61}$$

And ν_2 is obtained as

$$\nu_2 = \frac{\bar{K}}{\ln 2 \cdot \left(P_{\mathrm{RF,max}} + \phi_r \sum_{k=1}^{\bar{K}} 1/\left\|\tilde{\mathbf{g}}_k'\right\|^2\right)} - \kappa. \tag{62}$$

Algorithm 3 Solution of Problem (41)

Input: κ, ϕ_r, $\|\tilde{\mathbf{g}}_q\|^2$, $P_{\mathrm{RF,max}}$.

1: If (46b) is satisfied, go to step 2, otherwise go to step 3.

2: If (46a) is further satisfied, $p_q^\star$ is given as (45), otherwise $p_q^\star$ is given as (48), with ν_2 calculated by (49).

3: Obtain $\Theta = \{q | \ln 2 \cdot \kappa \phi_r < \|\tilde{\mathbf{g}}_q\|^2\}$. If (55) is satisfied, $p_q^\star$ is given as (54). Otherwise, find $\bar{K}$ based on (61), and then obtain ν_2 and $p_q^\star$ based on (62) and (59).

Output: $p_q^\star$.

Based on the above discussion, the process of solving (41) is concluded in Algorithm 3.

Next, we discuss the subproblem corresponding to $\{u_m\}$, which is formulated as

$$\begin{aligned} \max_{\{u_m\}} \ & \beta \log_2 \left(1 + \frac{A^2 R_{pd}^2 \left(\sum_{m=1}^M h_m \sqrt{u_m}\right)^2}{p_{\mathrm{clip}} R_{pd}^2 \left(\sum_{m=1}^M h_m \sqrt{u_m}\right)^2 + \sigma_v^2 \Delta f_v} \right) \\ & - \kappa \phi_v \sum_{m=1}^M u_m \\ \text{s.t.} \ & \phi_v \sum_{m=1}^M u_m \le P_{\mathrm{LED,max}} - P_{\mathrm{VLP}}^\star, \ 0 \le u_m \le \tau_m. \end{aligned} \tag{63}$$

The Lagrangian function of problem (63) is

$$\begin{aligned} & \mathcal{L}_3\left(\{u_m\}, \nu_3, \{\hat{\nu}_{3,m}\}, \{\tilde{\nu}_{3,m}\}\right) \\ & = \beta \log_2 \left(1 + \frac{A^2 R_{pd}^2 \left(\sum_{m=1}^M h_m \sqrt{u_m}\right)^2}{p_{\mathrm{clip}} R_{pd}^2 \left(\sum_{m=1}^M h_m \sqrt{u_m}\right)^2 + \sigma_v^2 \Delta f_v} \right) \\ & \quad - \kappa \phi_v \sum_{m=1}^M u_m - \nu_3 \left(\phi_v \sum_{m=1}^M u_m - P_{\mathrm{LED,max}} + P_{\mathrm{VLP}}^\star \right) \\ & \quad - \sum_{m=1}^M \hat{\nu}_{3,m}(u_m - \tau_m) + \sum_{m=1}^M \tilde{\nu}_{3,m} u_m. \end{aligned} \tag{64}$$

By letting $\partial \mathcal{L}_3 / \partial u_m = 0$, the optimal $u_m^\star$ is given as

$$\sqrt{u_m^\star} = \frac{\beta \psi^\star}{\ln 2\left[(\kappa + \nu_3)\phi_v + \hat{\nu}_{3,m}\right]} \cdot \tilde{h}_m \tag{65}$$

where

$$\psi^\star = \frac{\sum_{m=1}^M \tilde{h}_m \sqrt{u_m^\star}}{1 + \left(\sum_{m=1}^M \tilde{h}_m \sqrt{u_m^\star}\right)^2} - \sqrt{\frac{p_{\mathrm{clip}}}{A^2 + p_{\mathrm{clip}}}} \frac{\sum_{m=1}^M \bar{h}_m \sqrt{u_m^\star}}{1 + \left(\sum_{m=1}^M \bar{h}_m \sqrt{u_m^\star}\right)^2}$$

$\tilde{h}_m = \sqrt{(A^2 + p_{\mathrm{clip}}/\sigma_v^2 \Delta f_v) R_{pd}} h_m$, and $\bar{h}_m = \sqrt{(p_{\mathrm{clip}}/\sigma_v^2 \Delta f_v) R_{pd}} h_m$. The derivation of (65) is given in Appendix C.

Based on (65), if $u_m^\star = \tau_m$ (which means that $\hat{\nu}_{3,m} > 0$), we can derive that

$$\sqrt{\tau_m} < \frac{\beta \psi^\star}{\ln 2(\kappa + \nu_3)\phi_v} \cdot \tilde{h}_m. \tag{66}$$

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if $u_m^\star < \tau_m$ (which means that $\hat{\nu}_{3,m} = 0$), we have

$$\sqrt{u_m^\star} = \frac{\beta\psi^\star}{\ln 2(\kappa+\nu_3)\phi_v} \cdot \tilde{h}_m < \sqrt{\tau_m}. \tag{67}$$

According to (66) and (67), it can be concluded that

$$\sqrt{u_m^\star} = \begin{cases} \psi\tilde{h}_m, \text{ if } \sqrt{\tau_m}/\tilde{h}_m > \psi \\ \sqrt{\tau_m}, \text{ if } \sqrt{\tau_m}/\tilde{h}_m < \psi \end{cases} \tag{68}$$

where $\psi = (\beta\psi^\star/[\ln 2(\kappa+\nu_3)\phi_v])$.

Moreover, by sorting $\sqrt{\tau_m}/\tilde{h}_m$ in ascending order, i.e.,

$$\frac{\sqrt{\tau_{\omega_1}}}{\tilde{h}_{\omega_1}} \le \frac{\sqrt{\tau_{\omega_2}}}{\tilde{h}_{\omega_2}} \le \cdots \le \frac{\sqrt{\tau_{\omega_M}}}{\tilde{h}_{\omega_M}} \tag{69}$$

Equation (68) can be interpreted as

$$u_{\omega_m}^\star = \tau_{\omega_m}, \quad \text{for } \omega_1, \ldots, \omega_{\bar{M}} \tag{70a}$$

$$\frac{\sqrt{u_{\omega_{\bar{M}+1}}^\star}}{\tilde{h}_{\omega_{\bar{M}+1}}} = \frac{\sqrt{u_{\omega_{\bar{M}+2}}^\star}}{\tilde{h}_{\omega_{\bar{M}+2}}} = \ldots = \frac{\sqrt{u_{\omega_M}^\star}}{\tilde{h}_{\omega_M}} = \psi \tag{70b}$$

and thus (68) can be rephrased as

$$\sqrt{u_{\omega_m}^\star} = \min\left\{\sqrt{\tau_{\omega_m}}, \sqrt{u_{\omega_M}^\star}\tilde{h}_{\omega_m}/\tilde{h}_{\omega_M}\right\} \quad \text{for } \omega_1, \ldots, \omega_{M-1} \tag{71a}$$

$$\sqrt{u_{\omega_M}^\star} = \psi\tilde{h}_{\omega_M}. \tag{71b}$$

This indicates that we only need to find an appropriate $u_{\omega_M}^\star$, which can make the corresponding $\{u_{\omega_m}^\star\}_{m=1}^M$ given by (71) satisfy

$$\phi_v \sum_m u_{\omega_m}^\star = P_{\text{LED,max}} - P_{\text{VLP}}^\star, \quad \sqrt{u_{\omega_M}^\star} < \frac{\beta\psi^\star}{\ln 2 \cdot \kappa\phi_v}\tilde{h}_{\omega_M}$$

or

$$\phi_v \sum_m u_{\omega_m}^\star < P_{\text{LED,max}} - P_{\text{VLP}}^\star, \quad \sqrt{u_{\omega_M}^\star} = \frac{\beta\psi^\star}{\ln 2 \cdot \kappa\phi_v}\tilde{h}_{\omega_M}.$$

According to the above discussion, we conclude the process for solving problem (63) in Algorithm 4.

Algorithm 4 Solution of Problem (63)

Input: κ, ϕ_v, β, $\tilde{\mathbf{h}}$, $P_{VLP}^\star$, $P_{LED,\max}$ and a extremely small threshold δ.

1: sorting $\sqrt{\tau_m}/\tilde{h}_m$ in ascending order as shown in (69).
2: **Repeat**
3: set $u_{\omega_M}^{\min} = 0$ and $u_{\omega_M}^{\max} = \tau_{\omega_M}$.
4: update $u_{\omega_M}^\star = (u_{\omega_M}^{\min} + u_{\omega_M}^{\max})/2$ and obtain the reaming $u_{\omega_m}^\star$ with (71a)
5: If $\phi_v \sum_m u_{\omega_m}^\star \le P_{LED,\max} - P_{VLP}^\star$, set $u_{\omega_M}^{\min} = u_{\omega_M}^\star$. Otherwise, set $u_{\omega_M}^{\max} = u_{\omega_M}^\star$
6: **Until** $|\phi_v \sum_m u_{\omega_m}^\star + P_{VLP}^\star - P_{LED,\max}| < \delta$.
7: If $u_{\omega_M}^\star$ satisfies $\sqrt{u_{\omega_M}^\star} < \frac{\beta\psi^\star}{\ln 2\cdot\kappa\phi_v}\tilde{h}_{\omega_M}$, then end the Algorithm and output $\mathbf{u}^\star$. Otherwise, set $u_{\omega_M}^{\min} = 0$ and $u_{\omega_M}^{\max} = u_{\omega_M}^\star$, then got to step 8.
8: **Repeat**
9: update $u_{\omega_M}^\star = (u_{\omega_M}^{\min} + u_{\omega_M}^{\max})/2$ and obtain the reaming $u_{\omega_m}^\star$ with (71a).
10: If $\sqrt{u_{\omega_M}^\star} < \frac{\beta\psi^\star}{\ln 2\cdot\kappa\phi_v}\tilde{h}_{\omega_M}$, set $u_{\omega_M}^{\min} = u_{\omega_M}^\star$. Otherwise, set $u_{\omega_M}^{\max} = u_{\omega_M}^\star$.
11: **Until** $|\sqrt{u_{\omega_M}^\star} - \frac{\beta\psi^\star}{\ln 2\cdot\kappa\phi_v}\tilde{h}_{\omega_M}| < \delta$, output $\mathbf{u}^\star$.

V. Numerical Results

We consider a hybrid RF-VLCP system in a room of size $10 \times 10 \times 3$ m^3, which has $M = 4$ LEDs and an RF device with $N = 3$ antennas. The parameters of RF and VLC are listed in Tables I and II, respectively. The model of the RF channel $g_n(t)$ is same to [43], which is

$$g_n(t) = G_{\text{RF},n}\delta\left(t - \frac{d_{\text{RF},n}}{c}\right) = 10^{-\left(\frac{A\log_{10}(d_{\text{RF},n})}{10} + B + C\log_{10}\left(\frac{f_c}{5}\right) + X\right)}\delta\left(t - \frac{d_{\text{RF},n}}{c}\right). \tag{72}$$

In (72), $d_{\text{RF},n}$ is the distance between the nth antenna and the user. f_c is the carrier frequency in GHz. The parameters A, B, C are constants determined by the propagation model

$$\begin{cases} A = 18.7, \quad B = 46.8, \quad C = 20, \quad \text{for LOS case} \\ A = 36.8, \quad B = 43.8, \quad C = 20, \quad \text{for NLOS case} \end{cases} \tag{73}$$

and X is a constant related to the number of walls n_w. For light walls, $X = 5(n_w - 1)$. For heavy walls, $X = 12(n_w - 1)$. In this article, we consider the LOS case and set $n_w = 0$.

TABLE I
Parameter Settings of the VLCP System

PD responsivity R_p	0.4
Lambertian emission order ϖ	1
Detector area of PD receiver S	1 cm^2
DC bias I_{DC}	0.5 A
Signal bandwidth	20 MHz
Number of subcarriers K	128
Drain efficiency of PA μ_v	0.4
Maximum transmit power $P_{LED,max}$	1 W
Coordinates of LEDs l_t^m	[0; 3; 3], [−3; 0; 3] [3; 0; 3], [0; −3; 3]
PSD of noise σ_v^2	10^{-21} W/Hz
Circuit power $P_{c,RF}$	100 mWatt
Maximum transmit power $P_{LED,max}$	1 Watt

TABLE II
Parameters of the RF System

Carrier frequency f_c	5 GHz
Signal bandwidth	10 MHz
Number of subcarriers	64
Drain efficiency of PA μ_r	0.4
Maximum transmit power $P_{RF,max}$	1 W
Distance of RF antenna $d_{RF,n}$	6.35 m, 6.32 m, 6.29 m
Number of walls n_w	0
PSD of noise σ_r^2	3.98×10^{-21} W/Hz
Circuit power $P_{c,RF}$	100 mWatt
Maximum transmit power $P_{RF,max}$	1 Watt

All the numerical examples in this section are processed with MATLAB 2016B version, performing on a standard PC (CPU AMD Ryzen 5 4600H 3.0 GHz and 16 GB of RAM).

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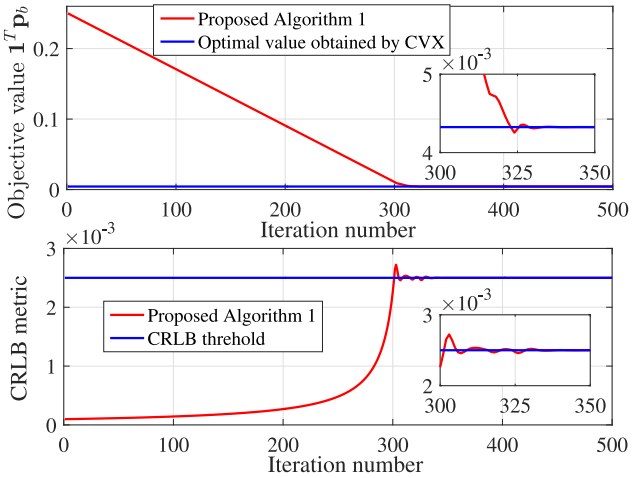

Fig. 3. Objective value (top) and CRLB metric (bottom) of problem (24) versus the iteration number, with CRLB threshold $\sqrt{\varepsilon} = 5$ cm.

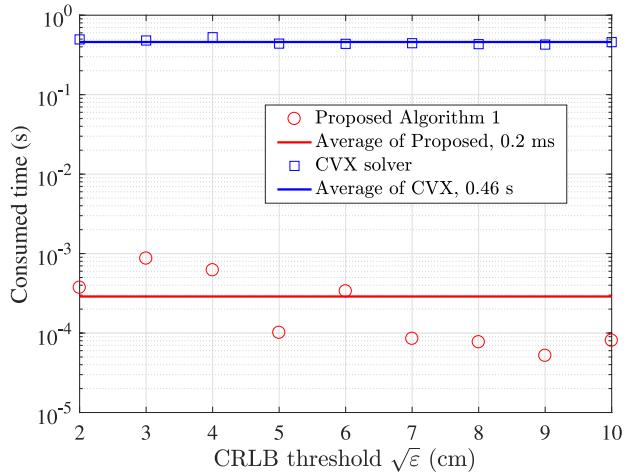

Fig. 4. Time consumed by Algorithm 1 (the first step of the proposed two-step algorithm), versus different CRLB thresholds. The consumption time of CVX solver is given as a benchmark.

A. Convergence Performance of the Proposed Algorithm

As a two-step method is proposed for tackling the joint optimization problem (21), let us start by discussing the performance of Algorithm 1 in the first step, which solves problem (24) to determine the power of the VLP signal. The CRLB threshold is set as $\sqrt{\varepsilon} = 5$ cm. In Fig. 3, we depict the iteration curves of objective value $\mathbf{1}^{\mathrm{T}}\mathbf{p}_b$ and the constrained CRLB metric. The top subfigure indicates that the objective value finally achieved by Algorithm 1 is same to that obtained by the CVX solver, while the bottom subfigure shows that the CRLB metric can be successfully constrained by the corresponding threshold. In addition, we also compare the computation time of Algorithm 1 and CVX solver in Fig. 4, where the consumed time of Algorithm 1 is recorded for achieving the convergence criterion (37). According to Fig. 4, the average time taken by the CVX solver is about 0.46 s, while that consumed by Algorithm 1 is only 0.2 ms. As a conclusion, the proposed Algorithm 1 takes much less time than the CVX solver to find the globally optimal solution of problem (21).

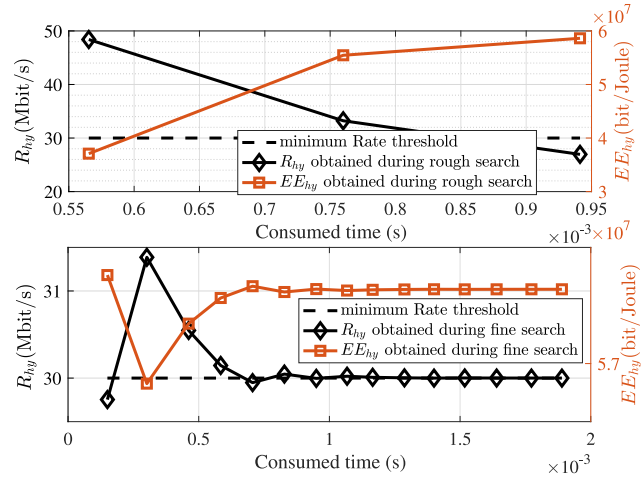

Fig. 5. Obtained R_{hy} and EE_{hy} during the iteration of Algorithm 2 (the second step of the proposed two-step algorithm). The top and bottom subfigures correspond to rough search and fine search, respectively.

Next, we discuss the second step of the proposed algorithm, which determines the communication related variables ($\mathbf{a}$ and $\mathbf{v}_q$) by solving problem (40) with Algorithm 2. The minimum data rate threshold is set as $R_{\min} = 30$ Mb/s and the normalized clipping level is set as $t_0 = 5$. Fig. 5 depicts the obtained R_{hy} and EE_{hy} during rough search and fine search. From the top subfigure, it can be observed that the update process of rough search makes R_{hy} decrease until $R_{hy} < R_{\min}$ is satisfied, and the total time taken by this process is less than 1 ms. Then, as shown in the bottom subfigure, the fine search further finds the exact $\mathbf{a}^\star$ and $\{\mathbf{v}_q^\star\}$ that make $R_{hy}^\star = R_{\min}$, which takes about 1.8 ms. Besides, it can be observed that the curve of EE_{hy} shows an opposite trend from the curve of R_{hy}. It first gradually increases during the rough search and then oscillates during the fine search until the optimal value is found.

To further illustrate the advantages of the proposed algorithm in terms of computation speed, we next compare it with the DSAC algorithm [29] in Figs. 6 and 7. We note that the setting of $t_0 = 5$ in this experiment stands for a large clipping level case, thus, the DSAC algorithm can find a good enough solution for problem (21). Fig. 6 shows that the proposed two-step algorithm obtains a slightly lower EE_{hy} than the DSAC algorithm. On the other hand, Fig. 7 indicates that the proposed algorithm requires much less computation time than the DSAC algorithm. By shortening the calculation time, the error introduced by using the estimated value $\hat{\boldsymbol{l}}_r$ can be reduced. Therefore, compered to the DSAC algorithm, the proposed algorithm is much more reliable in guaranteeing the positioning accuracy, while the degradation of the EE performance is almost negligible.

B. Performance of the Hybrid RF-VLCP System

In the following, we first discuss the impact of CRLB threshold ε on the hybrid RF-VLCP system, with $R_{\min}$ fixed as 30 Mb/s and t_0 set as 5. Fig. 8(a) uses $2b_m$ and $t_0\sqrt{2|\mathcal{K}_D|}a_m$ to reflect the power allocation results of VLP and VLC signals with different ε values. According to the figure, the

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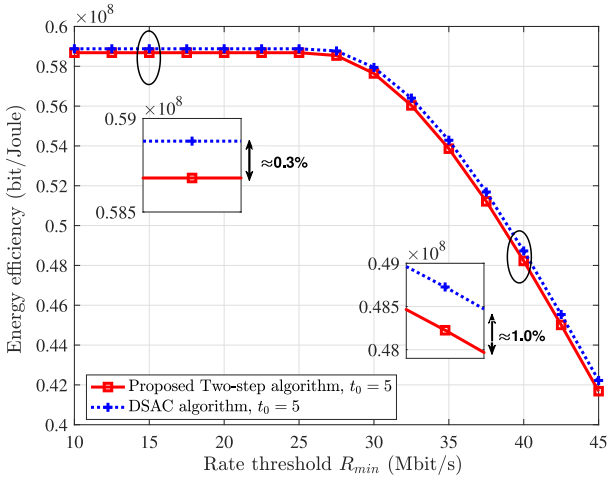


Fig. 6. Comparison of EE_{hy} obtained by the proposed two-step algorithm and the DSAC algorithm, with CRLB threshold $\sqrt{\varepsilon} = 5$ cm.

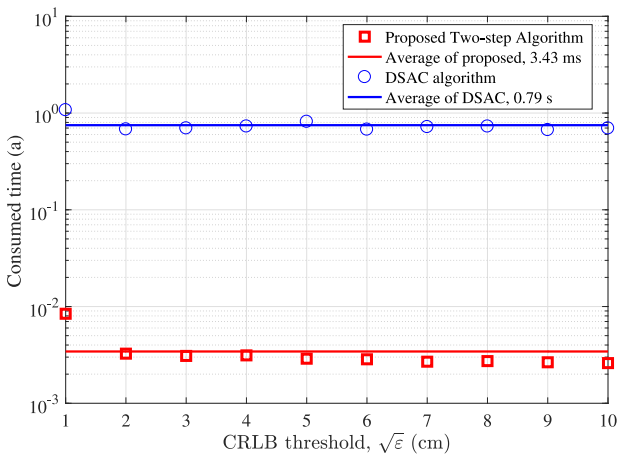
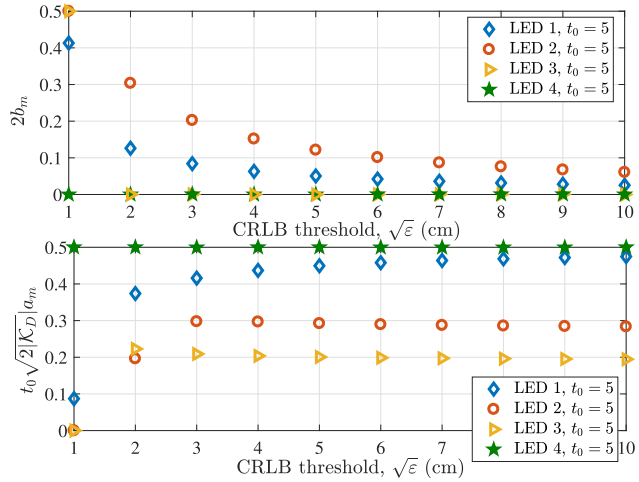
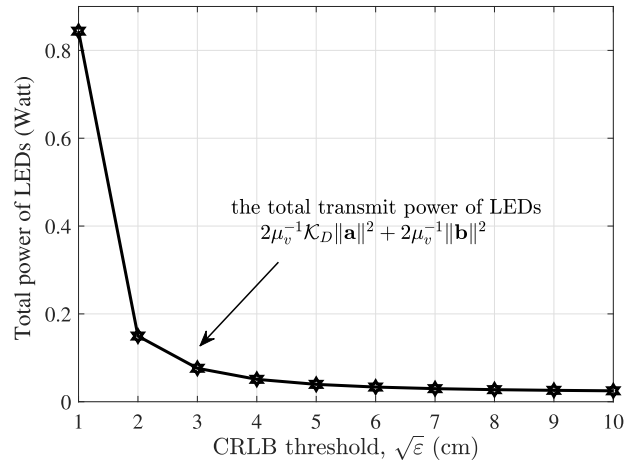


Fig. 7. Comparison of computation time of the proposed two-step algorithm and the DSAC algorithm, with data rate threshold $R_{\min} = 35$ Mb/s.



(a)



(b)

Fig. 8. Power allocation results of the VLP and VLC signals with different ε values. (a) Power allocation results of VLP and VLC signals. (b) Total transmit power of LEDs, with different CRLB values.

power required by VLP will decrease as the CRLB threshold ε becomes larger, and consequently, more power will be available for the VLC signals. It can also be verified from the figure that the nonnegativity constraint (5) is satisfied. Besides, Fig. 8(b) indicates that the designed $\{a_m\}$ and $\{b_m\}$ also satisfy the total power constraint, and the total transmitted power of LEDs will decrease when the CRLB threshold ε becomes larger.

Fig. 9 depicts the data rate provided by the VLC and RF links under different CRLB requirements. The green line shows that the VLC link can contribute more data rate as ε becomes larger because the power available for the link increases. The data rate provided by the RF link decreases accordingly since the total data rate requirement is fixed, which also indicates a reduction on the power consumption of the RF link. Fig. 10 depicts the EE performance of the hybrid RF-VLCP system with different ε values. It can be seen that the EE_{hy} metric improves as ε increases. This result is consistent with expectations, since we have shown that the power consumption of VLCP and RF reduces with the increase of ε, while the total data rate they provide remains unchanged. In addition, we further compare the EE performance of the hybrid RF-VLCP system with an RF and VLP system. In RF and VLP system, the data rate is entirely provided by the RF link, and the LEDs are only responsible for the positioning service. Thus, the EE metric for such a system is calculated as $R_{\mathrm{RF}}/(P_{\mathrm{RF}} + P_{\mathrm{VLP}} + P_c)$. According to Fig. 10, the EE performance of the hybrid RF-VLCP system is better than that of the RF and VLP system, verifying that using two links for communication is more energy efficient than using the RF link only.

Next, in Fig. 11, we evaluate the EE performance under different minimum data rate thresholds $R_{\min}$, with the CRLB threshold fixed as $\sqrt{\varepsilon} = 5$ cm. The figure shows that the EE performance first keeps as a constant and then decreases as $R_{\min}$ increases. This is because that the data rate requirement constraint is inactive for small $R_{\min}$. However, when $R_{\min}$ increases to a certain level, more transmit power is required to meet the data rate requirement, which leads to an overall decrease in the EE performance. Besides, the clipping level can also affect the EE performance. According to Fig. 12,

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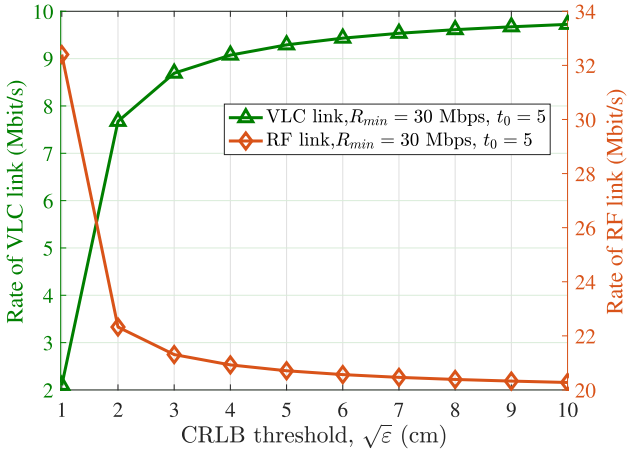

Fig. 9. Data rates provided by the VLC link and RF link, with different CRLB thresholds.

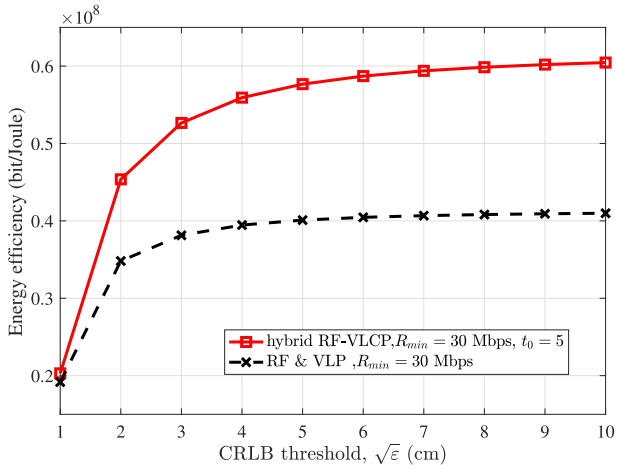

Fig. 10. Comparison of EE performance of hybrid RF-VLCP system and RF and VLP system under different CRLB thresholds. The former employs RF link and VLC link to jointly support the communication service while the latter only uses the RF link.

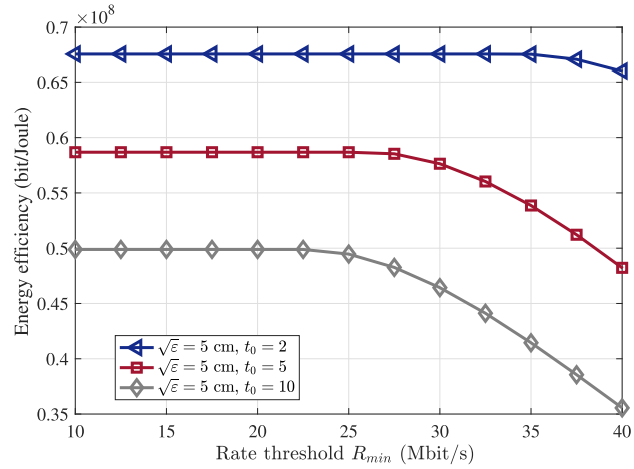

Fig. 11. EE performance of the hybrid RF-VLCP system with different minimum data rate thresholds.

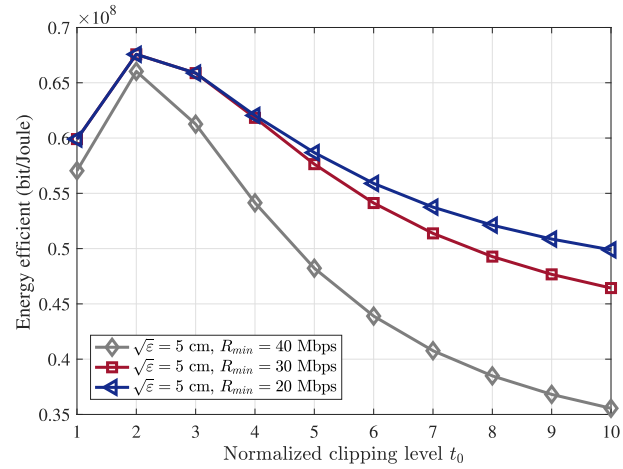

Fig. 12. EE performance of the hybrid RF-VLCP system with different clipping levels.

the EE performance is first improved as t_0 becomes larger, because a larger t_0 corresponds to a smaller attenuation degree of frequency domain signal and a lower clipping noise level. However, as t_0 keeps increasing, the feasible domain range of the nonnegativity constraint will be severely limited, thus leading to a worse optimization result for the EE_{hy} metric.

VI. Conclusion

In this article, we investigated an indoor hybrid RF-VLCP system, which provides positioning service through VLP technology and implements DL communication with VLC and RF links. The proposed hybrid system is able to provide high rate communication and high accuracy positioning simultaneously, thus filling the deficiency of the existing VLCP system and RF-VLC system. Besides, considering the single-user scenario, we proposed a joint beamformer design and power allocation method for the hybrid RF-VLCP system, which maximizes the EE while guaranteeing the qualities of positioning and communication.

In simulation part, we first demonstrated the advantages of the proposed algorithm in terms of computational efficiency. Then, the EE performance of the considered hybrid system was evaluated. We showed that the EE performance is improved with the relaxation of positioning error constraint and minimum data rate constraint. Besides, the clipping operation on the VLC signal can also affect the EE performance. As the normalized clipping level t_0 increases, the EE of the hybrid system first improves and then deteriorates. Furthermore, the proposed hybrid system was compared with an RF and VLP system, and the former was shown to be more energy efficient than the latter.

For future work, multiuser scenarios can be considered for the proposed hybrid RF-VLCP system, which is more realistic than single-user scenarios.

Appendix A

Reviewing the definition of Γ, we can derive

$$(\mathbf{I}_2 \otimes \mathbf{p}_b)^{\mathrm{T}}\Gamma = \begin{pmatrix} \sum_{m=1}^{M} \mathbf{p}_b[m]\chi_1^m\chi_1^m & \sum_{m=1}^{M} \mathbf{p}_b[m]\chi_1^m\chi_2^m \\ \sum_{m=1}^{M} \mathbf{p}_b[m]\chi_2^m\chi_1^m & \sum_{m=1}^{M} \mathbf{p}_b[m]\chi_2^m\chi_2^m. \end{pmatrix}$$

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Thus, we have

$$\begin{aligned}&\operatorname{trace}\left\{\left[\left(\mathbf{I}_2\otimes\mathbf{p}_b\right)^{\mathrm{T}}\Gamma\right]^{-1}\right\}\\&=\frac{\sum_{m=1}^{M}\mathbf{p}_b[m]\chi_1^m\chi_1^m+\sum_{m=1}^{M}\mathbf{p}_b[m]\chi_2^m\chi_2^m}{\sum_{m=1}^{M}\sum_{n=1}^{M}\mathbf{p}_b[m]\mathbf{p}_b[n]\left(\chi_1^m\chi_1^m\chi_2^n\chi_2^n-\chi_1^m\chi_2^m\chi_2^n\chi_1^n\right)}\\&=\frac{\sum_{m=1}^{M}\mathbf{p}_b[m]\left(\left(\chi_1^m\right)^2+\left(\chi_2^m\right)^2\right)}{\frac{1}{2}\sum_{m=1}^{M}\sum_{n=1}^{M}\mathbf{p}_b[n]\mathbf{p}_b[m]\left(\chi_1^n\chi_2^m-\chi_2^n\chi_1^m\right)^2}.\end{aligned}\tag{74}$$

Appendix B

With Algorithm 2, we find an $\kappa^\star$ such that the optimal solution of $\mathcal{P}_{\kappa^\star}$, i.e., $\{p_q^\star, u_m^\star\}$, also satisfies the rate constraint (40b). Now, we prove that $\{p_q^\star, u_m^\star\}$ is also the optimal solution of problem (40).

We first give the Lagrangian function of $\mathcal{P}_\kappa$ as follows:

$$\begin{aligned}\mathcal{L}_\kappa=&\sum_q\log_2\left(1+p_q\|\tilde{\mathbf{g}}_q\|^2\right)+\beta\log_2\left(1+\left(\sum_m\sqrt{u_m}\tilde{h}_m\right)^2\right)\\&-\kappa\left(\phi_r\sum_q p_q+\phi_v\sum_m u_m+P_{\mathrm{VLP}}^\star\right)\\&-\nu_2\left(\phi_r\sum_q p_q-P_{\mathrm{RF,max}}\right)\\&+\sum_q\tilde{\nu}_{2,q}p_q-\kappa\phi_v\sum_m u_m+\sum_m\hat{\nu}_{3,m}(u_m-\tau_m)\\&-\nu_3\left(\phi_v\sum_m u_m-P_{\mathrm{LED,max}}+P_{\mathrm{VLP}}^\star\right)+\sum_m\tilde{\nu}_{3,m}u_m.\end{aligned}$$

Since $\{p_q^\star, u_m^\star\}$ satisfies the Karush–Kuhn–Tucker (KKT) conditions of $\mathcal{P}_{\kappa^\star}$ and the rate constraint (40), we have

$$\frac{1}{\ln 2}\cdot\frac{\|\tilde{\mathbf{g}}_q\|^2}{1+p_q^\star\|\tilde{\mathbf{g}}_q\|^2}=\kappa^\star\phi_r+\nu_2\phi_r-\tilde{\nu}_{2,q}\tag{75a}$$

$$\frac{\beta}{\ln 2}\cdot\frac{\sum_m\sqrt{u_m^\star}\tilde{h}_m}{1+\left(\sum_m\sqrt{u_m^\star}\tilde{h}_m\right)^2}\frac{\tilde{h}_m}{\sqrt{u_m^\star}}=\kappa^\star\phi_v+\nu_3\phi_v+\hat{\nu}_{3,m}-\tilde{\nu}_{3,m}\tag{75b}$$

$$\nu_2\left(\phi_r\sum_q p_q^\star-P_{\mathrm{RF,max}}\right)=0,\quad\tilde{\nu}_{2,q}p_q^\star=0\tag{75c}$$

$$\nu_3\left(\phi_v\sum_m u_m^\star-P_{\mathrm{LED,max}}+P_{\mathrm{VLP}}^\star\right)=0\tag{75d}$$

$$\hat{\nu}_{3,m}\left(u_m^\star-\tau_m\right)=0,\quad\tilde{\nu}_{3,m}u_m^\star=0\tag{75e}$$

(40b) and (40c). (75f)

On the other hand, the Lagrangian function of problem (40) can be expressed as

$$L_\kappa+\chi\left(\begin{array}{l}\beta\log_2\left(1+\left(\sum_m\sqrt{u_m}\tilde{h}_m\right)^2\right)\\+\sum_q\log_2\left(1+p_q\|\tilde{\mathbf{g}}_q\|^2\right)-R_{\min}\end{array}\right).\tag{76}$$

By defining $\kappa^\circ=(\chi+1)\tilde{\kappa}$, $\nu_2^\circ=(\chi+1)\nu_2$, $\tilde{\nu}_{2,q}^\circ=(\chi+1)\tilde{\nu}_{2,q}$, $\nu_3^\circ=(\chi+1)\nu_3$, $\hat{\nu}_{3,m}^\circ=(\chi+1)\hat{\nu}_{3,m}$, $\tilde{\nu}_{3,m}^\circ=(\chi+1)\tilde{\nu}_{3,m}$, it can be easily checked that $\{p_q^\star, u_m^\star\}$ and the above new variables will satisfy the KKT conditions of problem (40). The proof is completed.

Appendix C

First, with the definitions of $\tilde{h}_m$ and $\bar{h}_m$, the Lagrangian function (64) can be further expressed as

$$\begin{aligned}&\mathcal{L}_3\left(\{u_m\},\nu_3,\{\hat{\nu}_{3,m}\},\{\tilde{\nu}_{3,m}\}\right)\\&=\beta\log_2\left(1+\left(\sum_{m=1}^{M}\tilde{h}_m\sqrt{u_m}\right)^2\right)\\&\quad-\beta\log_2\left(1+\left(\sum_{m=1}^{M}\bar{h}_m\sqrt{u_m}\right)^2\right)\\&\quad-\kappa\phi_v\sum_{m=1}^{M}u_m-\nu_3\left(\phi_v\sum_{m=1}^{M}u_m-P_{\mathrm{LED,max}}+P_{VLP}^\star\right)\\&\quad-\sum_{m=1}^{M}\hat{\nu}_{3,m}(u_m-\tau_m)+\sum_{m=1}^{M}\tilde{\nu}_{3,m}u_m.\end{aligned}\tag{77}$$

Thus, by letting $\partial\mathcal{L}_3/\partial u_m=0$, we have

$$\begin{aligned}&\frac{\beta}{\ln 2}\left(\frac{\left(\sum_{m=1}^{M}\tilde{h}_m\sqrt{u_m}\right)}{1+\left(\sum_{m=1}^{M}\tilde{h}_m\sqrt{u_m}\right)^2}\tilde{h}_m-\frac{\left(\sum_{m=1}^{M}\bar{h}_m\sqrt{u_m}\right)}{1+\left(\sum_{m=1}^{M}\bar{h}_m\sqrt{u_m}\right)^2}\bar{h}_m\right)\\&\quad\times\frac{1}{\sqrt{u_m}}=\phi_v(\kappa+\nu_3)+\hat{\nu}_{3,m}-\tilde{\nu}_{3,m}.\end{aligned}\tag{78}$$

To derive the explicit expression of $u_m^\star$ based on (78), we first denote that $\tilde{\nu}_{3,m}=0\quad\forall m$. If there exists $\tilde{\nu}_{3,m}\neq 0$, then, the corresponding $u_m=0$, which makes the left side of (78) infinite, thus contradicting the fact that $\{\nu_3,\hat{\nu}_{3,m}\}$ should be finite. Therefore, we have that $u_m^\star>0\quad\forall m$. Then, given that $\bar{h}_m=\sqrt{(p_{\mathrm{clip}}/[A^2+p_{\mathrm{clip}}])}\tilde{h}_m$ is established, (65) can be readily derived based on (78).

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